From Static to Smart: Designing a Planner-Centric Tool for Inventory Replenishment Policy Evaluation and Parameter Optimization

by

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Submitted to the Zaragoza Logistics Center in Partial Fulfillment of the Requirements for the Degree of

MASTER OF ENGINEERING IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT
in the

MIT-ZARAGOZA INTERNATIONAL LOGISTICS PROGRAM
at the

ZARAGOZA LOGISTICS CENTER,
A RESEARCH INSTITUTE ASSOCIATED WITH THE UNIVERSITY OF ZARAGOZA

May 2026

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Abstract

Specialty chemicals companies distributing high-value product families across global networks face a persistent inventory management challenge: the replenishment parameters embedded in Advanced Planning Systems are often static by design, yet the demand they must serve is heterogeneous and intermittent across destination markets. Over time, this mismatch produces inventory imbalances, deteriorating service levels, and growing volumes of manual planner overrides. This thesis studies that problem in the context of a specialty chemicals multinational, focusing on a single product family produced at a single central site and distributed to regional warehouses worldwide.

The developed solution is a modular Python-based decision-support tool that enables demand and supply planners to benchmark replenishment policies against one another under identical historical demand conditions. Demand forecasting is integrated into the framework not as a standalone objective but as a direct enabler of inventory decisions: a multi-layer valuation selects the best-performing demand forecasting and inventory model per SKU–Location combination, and its outputs feed directly into a policy simulation. Policy parameters are optimized for minimum average on-hand inventory at given service levels.

Applied to 12 material codes over a three-year observation window, the framework demonstrates that replacing static safety stock settings with data–driven, policy–specific parameters, reduces average inventory holdings without sacrificing service level, and that forecast accuracy improvements translate directly into quantifiable reductions in safety stock requirements. The tool surfaces these trade-offs through interpretable KPI dashboards and exports decision-ready parameter configurations for use in S&OP planning cycles, positioning the planner as the ultimate decision authority rather than a passive recipient of system recommendations.

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List of Abbreviations

ADI	Average Demand Interval	**MAE**	Mean Absolute Error
AIC	Akaike Information Criterion	**MAPE**	Mean Absolute Percentage Error
APO	Advanced Planning and Optimization	**MTO**	Make-To-Order
APS	Advanced Planning System	**MTS**	Make-To-Stock
CV	Coefficient of Variation	**PDF**	Probability Density Function
DES	Discrete-Event Simulation	**PPC**	Product-Plant Combination
DSS	Decision Support System	**RMSE**	Root Mean Square Error
EOQ	Economic Order Quantity	**ROP**	Reorder Point
ERP	Enterprise Resource Planning	**SARIMA**	Seasonal Autoregressive Integrated Moving Average
ETS	Exponential Smoothing State-Space Model	**SBA**	Syntetos-Boylan Approximation
GBM	Gradient Boosting Machine	**SKU**	Stock Keeping Unit
IP	Inventory Position	**S&OP**	Sales and Operations Planning
KPI	Key Performance Indicator	**SS**	Safety Stock
LT	Lead Time	**wSAA**	Weighted Sample Average Approximation
LSTM	Long Short-Term Memory		

1. Introduction

1.1. Industry Context and Motivation

Specialty chemicals companies operate global supply chains of considerable complexity, distributing a relatively small number of high-value product families across a large network of regional warehouses and end customers. Unlike fast-moving consumer goods, specialty chemical products are characterized by heterogeneous demand patterns across destinations, long and variable lead times linked to campaign-based production, and a customer base whose order behavior is sensitive to project cycles and seasonal application windows. These features place pressure on inventory management: holding too little stock risks, losing sales and damaging long-term customer relationships, while holding too much ties up working capital in products that may have limited shelf lives and high unit values.

Managing this complexity requires integrated digital planning systems. Enterprise Resource Planning (ERP) platforms provide the transactional backbone for procurement, production, and distribution, while Advanced Planning Systems (APS) extend this functionality with specialized modules for demand planning, supply planning, and inventory parameter management. Within such environments, replenishment decisions are typically governed by rule-based configurations embedded in the APS: reorder points, safety stock levels, lot sizes, and review periods that are set centrally and applied uniformly across SKU–location combinations. These configurations can perform adequately when their underlying assumptions remain valid, but they tend to degrade when demand behavior becomes more heterogeneous, network structures evolve, or service requirements tighten over time.

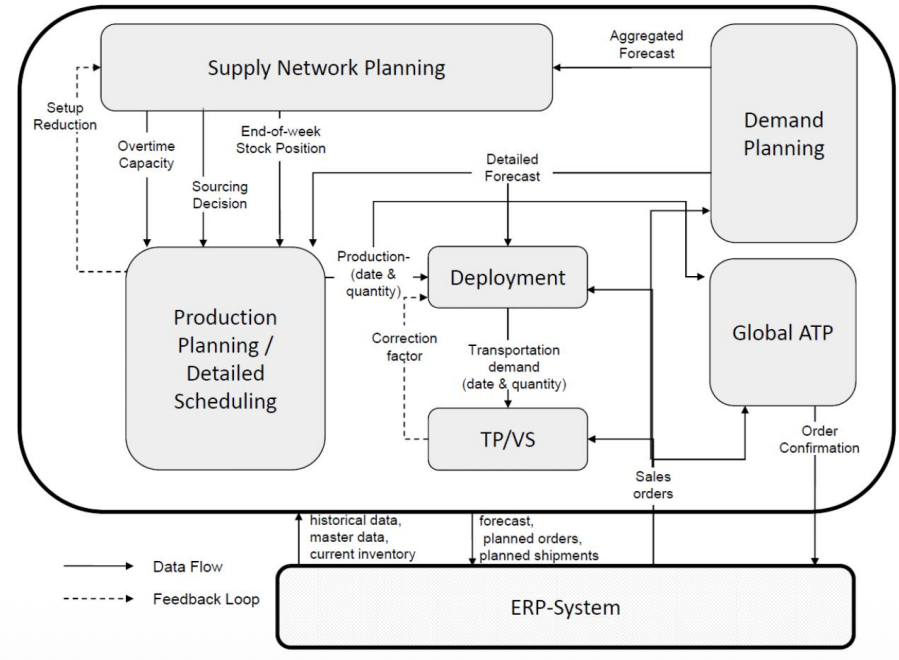


Figure 1. SAP APO block diagram. (Source: created by the authors)

Figure 1 illustrates the planning architecture of SAP Advanced Planning and Optimization (APO), the system used by the sponsor company. Demand forecasts generated by the sales department enter the system as inputs to both supply planning and deployment decisions. Inventory replenishment for make-to-stock products is governed by parameterized rules embedded in the system: fixed safety stock levels, reorder points derived from historical averages, and lot sizes linked to production campaign logic. This architecture is representative of a broad class of APS implementations in which replenishment logic is static by design, with parameters updated infrequently and without a structured feedback mechanism connecting realized demand performance to parameter revisions.

1.2. Problem Definition

The replenishment process for the product family under study is characterized by three structural limitations. First, the demand signal entering the planning system originates from the sales department in the form of a four-month rolling forecast, with no formal mechanism to validate it against realized shipment patterns or to correct systematic bias over time. Second, safety stock levels are fixed and reviewed rarely, leaving them unable to adapt to shifts in demand variability, lead time performance, or service level requirements across the global distribution network. Third, no structured framework exists for evaluating the current replenishment logic against alternative approaches. Planners lack a data-driven basis for determining whether the existing parameterization performs efficiently relative to other feasible policies.

These limitations produce a set of interrelated operational consequences. Deviations between the sales forecast and realized demand generate inventory imbalances across distribution destinations. Fixed safety stock settings offer insufficient protection during demand spikes or elevated lead time variability, while simultaneously inflating carrying costs during stable periods. Over time, the absence of any performance feedback erodes confidence in the planning system, resulting in increasing volumes of manual overrides and inconsistent replenishment decisions across production sites and regional warehouses.

At its core, the gap is structural: there is no feedback loop between replenishment outcomes and parameter design. Forecast quality determines the effectiveness of all downstream inventory decisions, yet neither forecast accuracy nor policy performance are measured systematically, and no analytical tool exists to evaluate whether alternative policies would improve the service–cost trade-off under real network demand conditions.

This thesis addresses that gap by developing a modular Python-based decision-support tool that enables demand planners to benchmark competing inventory policies against a live SAP APO baseline. The solution is developed for a single product family of a specialty chemical manufacturer. The scope covers 12 material codes produced in a single manufacturing site and shipped to destination plants across multiple regions. The scope is deliberately bound to balance analytical depth with practical usability for demand and inventory planners.

The remainder of this thesis is organized as follows. Chapter 2 reviews the academic literature that is relevant to the problem. Chapter 3 describes the methodological approach, while Chapter 4 presents the decision-support tool in full detail. Chapter 5 concludes with findings, recommendations, and directions for future work.

2. Literature Review

This chapter reviews academic literature across two streams directly relevant to the problem: demand forecasting methods for heterogeneous industrial portfolios and inventory management design, mainly through the support of simulation-based techniques. The discussion in both streams is structured to highlight the findings and frameworks in operations management to inform the design choices made in this thesis.

2.1. Demand Forecasting

Every replenishment decision in a supply chain begins with a question: how much will customers demand, and when? In this regard, the dominant paradigm has historically been time-series methods, which exploit the serial correlation structure of historical demand sequences to produce forward-looking estimates; working on the premise that past patterns carry enough signal to inform future decisions. Hyndman & Athanasopoulos (2021) provide a comprehensive review of time series methods, covering exponential smoothing state-space models (Hyndman et al., 2008), ARIMA-class models (Box et al., 1994), and the Holt-Winters seasonal extension. These methods share the property of relying exclusively on the univariate history of a single series and are well-understood in terms of their assumptions, limiting conditions, and failure modes.

For intermittent demand, the decomposition of inter-demand interval and demand size originally proposed by Croston (1972), corrected for bias by the Syntetos-Boylan Approximation (Boylan & Syntetos, 2021), consistently outperforms classical methods on slow-moving SKUs.

In multi-product forecasting settings, products with limited individual history can benefit from cross-series learning, whereby information from products with longer or richer demand histories is transferred across the portfolio. LightGBM, a gradient-boosted tree model, is particularly suitable for this context because it can pool information across SKU-location combinations while preserving combination-specific demand signals (Ke et al., 2017). Its strong empirical performance in large-scale forecasting applications, including the M5 forecasting competition, supports its use as a robust machine-learning benchmark for demand forecasting (Makridakis et al., 2022).

Nevertheless, no single method dominates across all demand patterns (Makridakis et al., 2022). This finding directly motivates the framework adopted in this study. Table 1 shows the candidate forecasting methods considered in this thesis that are evaluated per SKU-location combination on a rolling out-of-sample basis. The candidate set is deliberately bounded by the practical constraints of the sponsor's tool evaluation timeline and the deployment requirements of a production-ready Python tool.

Table 1. Comparison of forecasting methods considered for the demand forecasting module. Source: Created by the authors, based on the references cited in the table.

Method	Approach	Deployment Difficulty	Key Advantages	Key References
Naïve	Uses the last observed value as	Very Low	Simplest possible baseline; useful for benchmarking.	Hyndman & Athanasopoulos (2021)

Method	Approach	Deployment Difficulty	Key Advantages	Key References
	the forecast for future periods.			
Moving Average	Averages the most recent observations to produce a forecast.	Very Low	Smooths short-term noise; straightforward to interpret.	Hyndman & Athanasopoulos (2021)
Exponential Smoothing (ETS)	Assigns exponentially declining weights to past observations.	Low	Adapts more rapidly than simple averages; handles level shifts.	Hyndman et al. (2008)
Holt-Winters	Extends exponential smoothing to capture trend and seasonality.	Medium	Well-suited to products with repeating seasonal patterns.	Hyndman & Athanasopoulos (2021)
Croston / SBA	Designed for series with intermittent demand and sporadic orders.	Medium	Outperforms standard time-series methods for slow-moving SKUs.	Croston (1972); Boylan & Syntetos (2021)
LightGBM (Gradient Boosting)	Machine-learning model using engineered lag features, rolling statistics, and cross-product patterns.	Medium	Robust to limited history; supports global training across many SKUs; demonstrated superiority in large-scale forecasting competitions.	Ke et al. (2017); Makridakis et al. (2022)

2.2. Inventory Policy Design and Simulation-Based Decision Support

The theoretical foundation for inventory management under demand uncertainty is well-established. Classical continuous-review and periodic-review policy families, including base-stock and order-up-to structures, provide analytical expressions for replenishment decisions under assumptions such as stationary demand and known lead-time distributions (Silver et al., 2021). These policy structures serve as the benchmark against which the simulation-based evaluation in this thesis is developed.

Applying these policies effectively, however, requires first distinguishing which products should be stocked at all. D' Alessandro & Baveja (2000), working in a specialty-chemicals manufacturing

context directly comparable to this thesis, show that assigning intermittent SKUs to make-to-order production can reduce speculative inventory without compromising availability. Their quadrant-based classification, using demand volume and variability as segmentation criteria, provides the direct template for the MTS/MTO filter applied in our work before any policy parameterization is attempted.

Once the MTS segment is isolated, the quality of replenishment decisions depends heavily on forecast accuracy. Babai et al. (2013) analyzing 329 SKUs across a two-stage retailer-manufacturer supply chain, confirm that reducing forecast error translates directly into lower safety stock requirements without sacrificing service levels. This finding is the theoretical basis for integrating forecasting directly into the policy simulation engine: improvements in forecast accuracy are expected to propagate into measurable reductions in required safety stock.

Building on this forecasting-inventory link, a recent stream of research has pushed toward prescriptive approaches that derive replenishment quantities directly from data, bypassing the conventional predict-then-optimize sequence entirely. Schmidt & Pibernik (2025) demonstrate that a global Random Forest model embedded in a rolling-horizon optimization framework can reduce total inventory costs by up to 38.9% across a heterogeneous pharmaceutical portfolio. Qi et al. (2023) go further, proposing an end-to-end deep learning framework that reduced total inventory costs by 40.4% in a live experiment at JD.com, while Madeka et al. (2022) present a complementary Deep Reinforcement Learning approach at Amazon scale that achieved a 12% reduction in inventory levels with no significant revenue loss. Boute et al. (2022), however, provide a more cautious assessment of this frontier noting that such approaches often remain restricted to stylized settings, require extensive hyperparameter tuning, and may suffer from training instability at realistic portfolio scales.

These results are promising, but they emerge from contexts with tens of thousands of SKUs, dedicated ML engineering teams, and substantial computational infrastructure. These conditions are rarely present in mid-sized industrial settings. In a specialty chemicals environment with 12 material codes and planners who require a transparent, interpretable tool for SOP planning cycles, a simulation-based approach is more appropriate.

Turban et al. (2011) define a decision-support system as a model base, database, and user interface designed to surface alternatives rather than prescribe answers. Badri (1999) operationalizes this in an inventory context: a multi-product simulation engine evaluates thousands of policy combinations and presents them as a comparison the manager interrogates rather than implements.

Following the decision-support design philosophy established by Turban et al. (2011) and demonstrated in an inventory context by Badri (1999), the framework developed here surfaces trade-offs through policy simulation and leaves the final replenishment decision to the planner.

3. Methodology

Our approach in studying the problem and designing the solution consists of multiple stages that move progressively from problem understanding to solution design (Figure 2. Thesis methodology chart. The process begins with analytical problem framing translating the sponsor company's operational planning challenge into a set of clearly defined research questions and design requirements. This is followed by a data preparation stage, in which raw historical demand and supply records are cleaned, validated, and structured to serve as inputs to the forecasting and simulation engines. The tool framework is then developed iteratively, an initial working interface is built and populated with cleaned data, enabling early-stage feedback and validation sessions with the sponsor. These feedback cycles ensure that the solution design including policy selection process, KPI definitions, and dashboard structure reflects both the academic rigor in the literature review and the practical requirements of the planners who will use the tool.

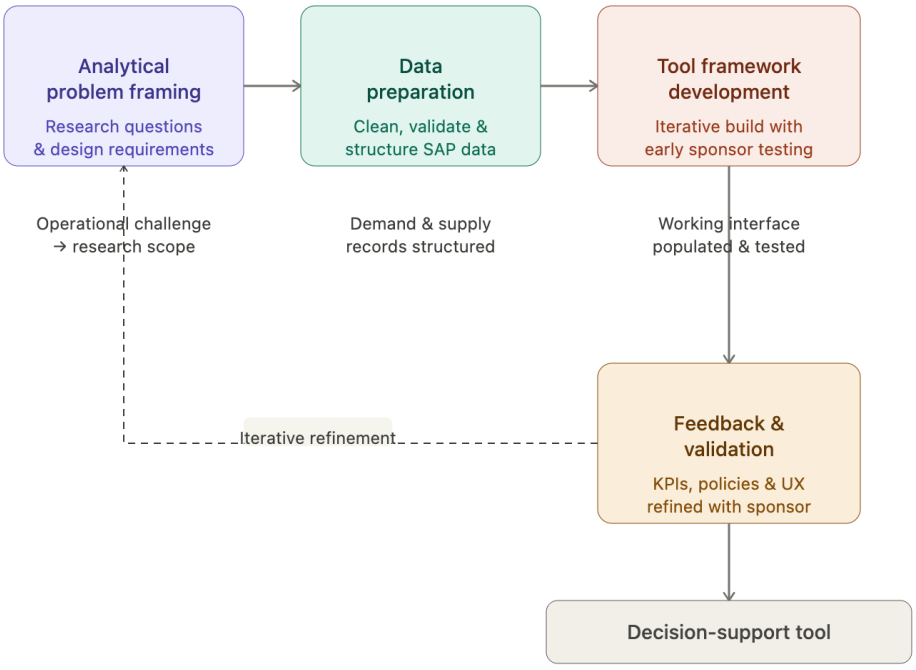


Figure 2. Thesis methodology chart.

3.1. Analytics Problem Framing

Inventory management in this supply chain operates under two fundamentally different replenishment logics, reflecting the distinct roles of each echelon. Tier 2 refers to the regional warehouses, where replenishment is shaped by local customer demand: each warehouse serves a specific customer base, and its consumption pattern should directly inform reorder decisions. Tier 1 refers to the production plants, where replenishment is shaped by campaign batch-sizing constraints, manufacturing lead times, and aggregated demand from downstream warehouses, rather than by direct response to individual customer orders. Applying a single uniform policy across both tiers would therefore be inappropriate. It could either oversize production batches relative to actual downstream pull or leave regional warehouses without sufficiently responsive replenishment coverage.

The analytical framework therefore maintains a deliberate distinction between Tier 2 and Tier 1 replenishment logic throughout the thesis.

Underlying both tiers is a shared dependency on forecast quality. Replenishment parameters, whether a reorder point, safety stock buffer, or order-up-to level are only as reliable as the demand signal they are built on. In the current planning process, the demand signal entering SAP APO originates from sales as a rolling forecast, with no formal mechanism to validate it against realized shipments or correct systematic bias over time. The analytical framework therefore treats forecasting not as a standalone objective but as a direct enabler of inventory performance: a dedicated forecasting evaluation is embedded as the first analytical stage, and its outputs feed directly into the policy simulation.

Within this two-tier structure, the thesis addresses three analytical questions: Which forecasting methods best capture the heterogeneous demand patterns in the portfolio? How do alternative replenishment policies compare against the current SAP APO baseline under identical historical demand conditions? And, which policy configurations deliver the best balance between inventory investment and service level, per SKU-location combination?

3.2. Data Preparation

The empirical analysis draws on three years of transactional SAP data from October 2022 to September 2025, covering outbound shipment movements, inbound batch receipts, opening inventory snapshots, and a SKU-level master data file. The unit of analysis is product-plant-combination (PPC).

Raw data preparation involves temporal alignment, material and plant code normalization across source files, and reconstruction of inventory trajectories from movement records. SKU-location combinations are then classified as MTS or MTO using the segmentation logic of D' Alessandro & Baveja (2000), with only MTS combinations entering the policy simulation. A consistent train-test split is applied across all forecast evaluations and policy simulations, so that reported performance metrics reflect behavior on genuinely unseen data. Full preprocessing details are documented in Appendix B: Data and Preprocessing.

3.3. Solution Architecture Overview

The solution is organized into three connected modules, each corresponding to a distinct stage of the decision-support pipeline.

The **Demand Forecasting Module** evaluates multiple candidate methods per SKU-location combination on a rolling out-of-sample basis and selects the best performer as the direct forecast input to the simulation engine.

The **Inventory Policy Simulation Module** replays historical demand through a set of named replenishment policies applied separately for Tier 1 and Tier 2 under identical demand conditions. For each policy, the simulation automatically evaluates alternative parameter values and selects the values that achieve the target service level with the lowest average inventory. The resulting recommendation serves as a data-grounded default that the planner can inspect and override before results are finalized.

The Planner **Decision-Support Interface** surfaces simulation results through interpretable KPI dashboards, allowing planners to compare policy performance across tiers and export decision-ready parameter configurations for use in SOP cycles without requiring any programming knowledge.

4. Forecasting and Inventory Optimization Solution

The three structural limitations identified in the introduction a biased demand signal, static safety stock parameters, and no policy benchmarking capability each produce a measurable operational cost that the current SAP APO setup cannot correct on its own. The following sections present evidence that the framework closes all three gaps through a deliberate two-stage evaluation that first challenges the current replenishment parameters within the existing planning horizon, then tests whether reasonable changes in the planning horizon unlock further gains. Every result is measured against the live SAP baseline the planner already works with.

4.1. Forecasting Model Selection and Performance Evaluation

Forecast accuracy is the foundation that both evaluation stages rest on. In Stage 1, where the planning horizon is held at the sponsor's current setting, the forecast directly sets the order quantity in the Forecast-Based policy and sizes the safety-stock buffer across all policies through the standard deviation of forecast errors on the test set. In Stage 2, where the horizon is freed as an optimization variable, forecast accuracy determines how far the horizon can be shortened before service deteriorates. Shorter horizons are less exposed to forecast error, but only if the underlying signal is reliable enough to support tight replenishment windows. The framework's contribution is therefore not a single best forecaster but a per-combination selection mechanism: every combination receives the method best suited to its demand profile, and both stages inherit that accuracy gain.

The solution evaluates four candidate forecasting models per product–plant combination on a fixed 24-month / 12-month train/test split and selects, per combination, the method that minimizes out-of-sample RMSE. The four candidates are Naïve Forecasting (last observed value, held flat), Moving Average (mean of the three most recent periods, held flat), Holt–Winters (triple exponential smoothing with additive trend and seasonality), and a global LightGBM model, a gradient-boosted tree model trained jointly across all eligible combinations. Naïve Forecasting and Moving Average serve as the minimum performance bar; Holt–Winters requires ≥ 24 months of training history and falls back to the historical mean otherwise. The LightGBM model is the central element of the forecasting solution and the only one that learns across combinations.

The following design choices in the LightGBM specification explain the observed accuracy gains on this dataset:

- **Direct multi-step strategy:** An initial recursive implementation, in which each prediction was fed back as the lag for the next step, collapsed toward the moving average within a few horizons because errors compounded. The current solution computes lag features once from observed history at reference time t and makes a separate prediction for each horizon $h \in \{1, \ldots, 24\}$ from the same fixed feature vector: only the horizon and target month change across steps. This eliminated error compounding and was the single largest driver of the improvement over the classical baselines on intermittent series.
- **Intermittency-aware feature set:** Nine features are used: lag_1, lag_2, lag_3, lag_12 (annual seasonality), rolling_mean_3, nonzero_frac_6 (the fraction of the last 6 months with positive demand), month_of_year, horizon, and a label-encoded combination identifier. The variable nonzero_frac_6 measures the fraction of the previous six months with positive demand. This feature is important because it allows the model to learn that a combination

with nonzero_frac_6 = 0.33 is likely to produce a near-zero forecast even when lag_1 was non-zero, a pattern that often breaks naive lag-based methods in intermittent-demand portfolios.

- **Additional implementation choices:** The model also uses walk-forward selection of n_estimators through 3-fold cross-validation and inverse-volume sample weighting, so that small-volume combinations are not drowned out by large ones in the RMSE objective. Together with the eligibility filter and fallback logic for unseen combinations, these choices are documented in Appendix C: LightGBM Global Model – Training Detail.

Whether these design choices translate into accuracy gains on the sponsor's portfolio is an empirical question. Table 2 and Table 3 report mean out-of-sample accuracy across all 21 PPCs and 11 materials (aggregated global demand) over the test period, respectively. Both views are reported because they expose different parts of the picture: the material-level view tests whether the global model captures total-network demand patterns, while the combination-level view tests whether it generalizes down to the granularity at which inventory decisions are actually made and at which both Stage 1 and Stage 2 parameters are optimized.

Table 2. Mean forecasting accuracy across all MTS product-plant combinations.

Forecasting Method (Results for **21 PPCs**)	**Weighted Avg. MAE (%)**	**Avg. RMSE (kg)**	**Avg. Bias (kg)**
Naive	5.6%	19,070	-9,643
Moving Average (3)	5.0%	16,943	-9,449
Holt-Winters	5.4%	18,171	-9,320
LightGBM (Global AI)	3.8%	12,992	-3,252

Table 3. Mean forecasting accuracy across all materials (global demand).

Forecasting Method (Results for **11 materials**)	**Weighted Avg. MAE (%)**	**Avg. RMSE (kg)**	**Avg. Bias (kg)**
Naive	4.2%	36,000	-18,426
Moving Average (3)	3.9%	34,118	-20,757
Holt-Winters	4.0%	34,555	-19,712
LightGBM (Global AI)	2.8%	23,833	-7,362

LightGBM achieves the lowest mean RMSE in both views, with the largest gains on combinations exhibiting short history and intermittent demand: the conditions the direct multi-step strategy and the intermittency feature were specifically designed to handle. The bias column reinforces the

same point: classical methods systematically under-forecast by 9,000–10,000 kg on average at combination level (and nearly twice that at material level), while LightGBM's bias is reduced by roughly 65–70%. Under-forecasting is the more costly direction operationally, since it propagates straight into stockouts through the reorder-point logic of Section 4.2.

The wins do not come from a single method dominating uniformly. At material level (aggregated global demand) LightGBM performs best on all 11 materials, but at combination level it leads on 15 of 21 PPCs (10 of 12 Tier 1). The remaining 6 PPCs split into two groups: on 4 of them Holt–Winters wins by a negligible margin, and on the remaining 2 the winning method is Naive on combinations whose share of zero-demand months exceeds 40%. This is consistent with Makridakis et al. (2022): no single method dominates across all demand patterns. The forecasting solution's value is therefore not just the global model itself but the per-combination evaluation framework around it: every combination receives the method best suited to its demand profile, and the planner sees the ranking that drove the choice. This accuracy foundation holds equally for the fixed-horizon results of Stage 1 and the freed-horizon results of Stage 2.

4.2. Inventory Policy Simulation and Parameter Optimization

Across the 21 MTS combinations evaluated, the framework produces two structurally different findings depending on tier. At Tier 2, automated parameter optimization reduces average inventory by 17%, raises in-stock from 83% to 91%, and cuts holding cost by 18%, improving all three dimensions simultaneously. At Tier 1, where the actual baseline already operates at 97.5% in-stock, no portfolio-wide policy improves on the baseline along both dimensions. Instead, the framework surfaces a cost-service trade-off the planner can interrogate per combination. The remainder of this section presents the evidence behind both findings, organized in two stages that progressively relax the constraints under which the planner operates.

Inventory optimization in this framework is a structured decision space rather than a fixed prescription. For each policy and each product-plant combination, the simulation engine systematically explores alternative parameter values, including reorder level, safety stock buffer, order quantity, and planning horizon. It then selects the parameter values that minimize average on-hand inventory, while meeting the target service level. The results are surfaced as editable defaults the planner can inspect, challenge, and override before any decision is finalized.

Table 4. Implemented inventory policies by tier. lists the inventory policies evaluated per tier under identical demand sequences and shared parameters (lead time, target service level, frozen horizon). The Forecast-Based Reorder-Point policy is included at both tiers as the natural baseline, because it reflects the sponsor's current SAP APO logic and enables a fair comparison under optimized parameters. The alternative at each tier is chosen to fit the tier's replenishment reality. At Tier 1, an EOQ-bounded continuous-review (s, S) policy aligns with campaign-production constraints. At Tier 2, an Order-Up-To (S) periodic-review policy maps onto the fixed inbound shipment cadence from Tier 1. This tier-aware policy library is what allows the comparison to be operationally meaningful. Each alternative is expressed in terms that the planner already uses, so a recommendation to switch policies translates into a change in working practice rather than into a modelling abstraction the planner must reinterpret.

Table 4. Implemented inventory policies by tier.

	Tier 1 - Production Plants	Tier 2 - Distribution Plants
Policy 1	Forecast-Based APO Reorder-Point	Forecast-Based APO Reorder-Point
Policy 2	(s, S): Continuous review, EOQ bounded order quantity	Order-Up-To (S): Periodic review

Policy 1 – Forecast-Based Reorder-Point. This policy uses the forecast both to trigger orders and to size them, reflecting the sponsor's SAP APO reorder-point logic. The projected stock level is monitored at every period, and an order is placed whenever it falls below a reorder point. The reorder point combines mean forecasted demand over the lead time with a safety-stock buffer sized from the standard deviation of forecast errors on the test set. In this way, the policy hedges against forecast inaccuracy rather than raw demand variability. Order quantities cover at least the forecasted demand over the frozen horizon and are rounded up to the nearest full pallet. The simulation advances through the test period in rolling steps, where the forecast horizon is itself a tuned parameter. For each combination, a grid search over candidate horizons selects the value that minimizes average inventory subject to the target service level. At every step, the model is re-fitted on the expanded training window, a fresh forecast is produced, and replenishment decisions are made against it. The frozen portion of each rolling window is treated as already committed, mirroring how a planner operates in practice.

Policy 2 at Tier 1 – EOQ-bounded (s, S). This is a continuous-review policy adapted to campaign production. Whenever the inventory position falls below the reorder level *s*, a campaign is initiated. The order quantity follows a two-case rule. When forecasted demand over the average production interval exceeds the EOQ, the campaign is capped at the EOQ to avoid producing a batch larger than the economically optimal run. When forecasted demand is below the EOQ, the campaign is sized to forecasted demand, avoiding a run that builds inventory beyond what near-term consumption will absorb. In both cases, the order quantity $Q = \min(D_f, EOQ)$, where D_f is the forecasted demand over the average production interval. EOQ values are sourced directly from the master-data file. The policy raises a warning whenever the frozen horizon exceeds the implied average production interval, since this would require back-to-back campaigns and is operationally infeasible.

Policy 2 at Tier 2 – Order-Up-To (S). This is a periodic-review policy, where inventory is reviewed every R periods and an order is placed to bring the inventory position up to a target level S. No orders are placed between review dates, regardless of the on-hand inventory level. The target S and the safety-stock buffer are computed from the best-fit demand distribution rather than from a fixed normality assumption. Candidate distributions (Normal, Poisson, Weibull, Exponential, and Gamma) are fitted to the training series and ranked by Akaike Information Criterion (AIC). This approach is more robust than assuming normal demand for combinations whose monthly demand is skewed or contains many zero-demand months, a pattern commonly observed in the Tier 2 portfolio.

The framework's findings are organized in two stages that progressively relax the constraints under which planners operate. **Stage 1** evaluates whether parameters can be improved without changing

the production planning horizon. This is the most immediately deployable form of the framework, since it requires no renegotiation of downstream commitments. **Stage 2** relaxes both the planning horizon and the policy choice, evaluating whether further gains are available when the planning architecture itself is allowed to change. The two stages answer different deployment questions and produce different headline findings.

4.2.1. Stage 1 – Fixed-Horizon Parameter Optimization

Stage 1 isolates the parameter-tuning problem from any change to the production planning architecture itself. The forecast horizon, the lead time, the safety-stock formula, the rounding rule, and the trigger logic are all held at their current SAP APO values; the only variable allowed to move is the reorder point. This is the form of improvement the planner can deploy without renegotiating production cadences downstream, and it is therefore the right starting point for evaluating whether a data-driven layer adds value before changing how the system operates. The contrast isolates a single question: does replacing the analytically derived reorder point with a value optimized against realized demand improve the cost–service trade-off without changing anything else in the planning system?

Two configurations of the forecast-based reorder-point policy are simulated under identical demand sequences and identical shared parameters (lead time, frozen horizon = 1 period, target service level = 95%). The first is the current SAP APO setup, in which the reorder point is computed analytically as forecasted demand over the lead time plus a safety-stock buffer, calculated using the standard deviation of forecast errors on the test set and z, the safety factor associated with the target service level.

The second is a data-driven replica (DDR) of the same policy: trigger logic, order-quantity formula, and pallet rounding are unchanged; only the reorder point is replaced with a value found by grid search on the training set, minimizing average on-hand inventory subject to a service-level floor equal to the actual in-stock percentage observed during training. The optimization is bounded below by the standard deviation of the forecast error, so that the reorder point cannot collapse below the buffer that variability in forecast accuracy alone would justify.

Tier 1 Results

Table 5 reports portfolio averages across the 11 Tier 1 MTS combinations evaluated over the 12-month test window. Three policies are compared: the actual on-hand position, the analytical Forecast-Based reorder point as currently configured in SAP APO, and the data-driven replica with the optimized reorder point.

Table 5. Tier 1 PPCs portfolio comparison – Stage 1 reorder-point optimization.

KPI	Actual	Forecast-Based (APO)	DDR Policy 1
Re-Order Point (kg)	-	65,369	34,549

KPI	Actual	Forecast-Based (APO)	DDR Policy 1
Portfolio avg. on-hand (kg)	139,829	195,506	174,215[1]
In-stock (%)	97.5%	99.2%	97.7%
Production campaigns / yr	3.07	3.49	2.70
Avg. order size (kg)	187,281	189,828	171,112
Est. annual holding cost (€)	48,365	67,263	59,717
Est. annual changeover cost (€)	71,774	81,542	63,068
Total cost (€)	120,139	148,805	122,784

Two findings emerge at the portfolio level. The first is the cost of the analytical reorder-point as currently parameterized: applied without modification across the eleven combinations, it carries 40% more inventory than the actual baseline and raises total annual cost from €120,139 to €148,805, a 24% increase. Service improves from 97.5% to 99.2%, but the marginal cost of those 1.7 percentage points is roughly €29,000 per year, paid for almost entirely through additional holding cost.

The second finding is what the optimization recovers: the data-driven replica reduces average on-hand inventory by 21,300 kg relative to the analytical APO setup (-11%), brings total annual cost back to €122,784 (within €2,600 of the actual baseline), and does so while still delivering 97.7% in-stock, marginally above the actual 97.5%. The improvement comes from both cost levers simultaneously: holding cost falls by €7,500 as the lower reorder point releases stock, and changeover cost falls by €18,500 as fewer triggers convert into 2.7 campaigns per year against the 3.5 produced by the over-buffered analytical setup.

Importantly, the gain is not produced at the expense of service. DDR Policy 1 reaches a service level within 0.2 percentage points of the actual operational baseline while taking €26,000 of annual cost out of the system, and the optimization is bounded so that it cannot drop the reorder point below the σ-floor implied by forecast-error variability. The result is consistent with what the literature anticipates: when forecast errors are small relative to the reorder-point buffer in use, the

[1] Average on-hand inventory is computed on a finite simulation horizon. A replenishment order placed late in the test window contributes its full received volume to the average even though most of it would be consumed after the window closes, producing a mild upward bias whose size depends on the policy's order-timing behavior rather than its steady-state inventory profile. The bias affects all simulated policies over the same horizon, so cross-policy comparisons within a tier remain informative; absolute figures should be read with this caveat in mind. The same caveat applies to the average-inventory KPI reported in every results table in this chapter.

buffer is the binding constraint and tightening it releases inventory without compromising service (Babai et al., 2013).

The portfolio average masks meaningful heterogeneity across combinations, which is informative because it indicates where the optimization is doing useful work and where it is not. Each of the eleven Tier 1 PPCs is classified into one of four scenarios according to whether DDR Policy 1 reduces total cost relative to the actual baseline and whether it maintains or improves service. The breakdown is shown in Figure 3.

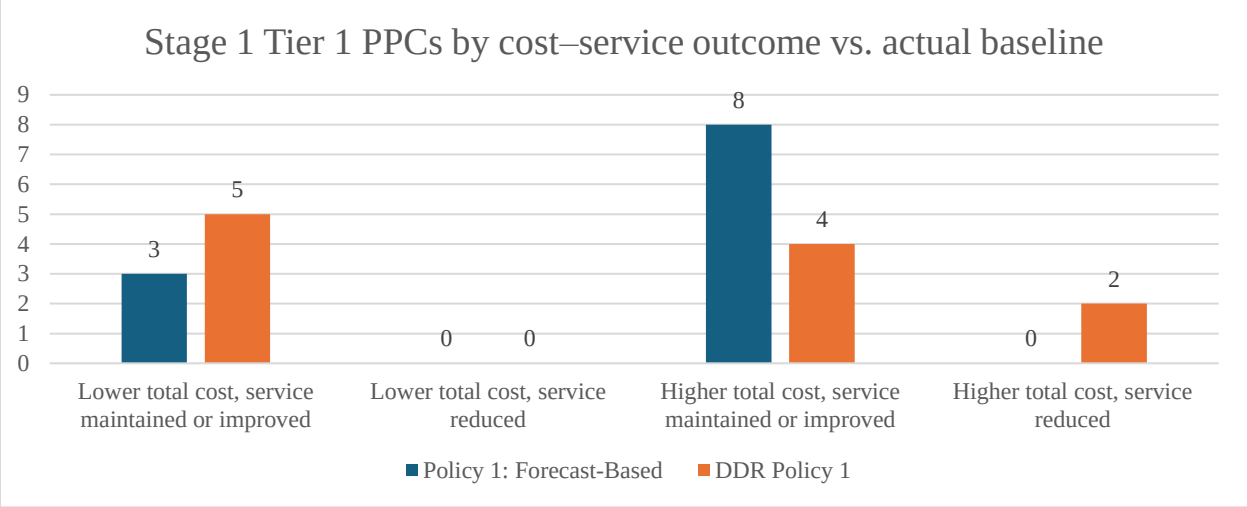


Figure 3. Distribution of the 11 Tier 1 PPCs across the four cost–service quadrants under each policy, benchmarked against the actual on-hand baseline.

The five combinations in the unambiguous-improvement quadrant are cases where the analytical reorder point was systematically over-buffered: the DDR optimization releases inventory and consolidates triggers into fewer campaigns without affecting fulfilment. Cost reductions in this group are driven by both levers simultaneously, with the largest single contribution coming from changeover cost as the campaign count falls. No combination in this quadrant has its service degraded by the optimization, confirming that the gains are not extracted by under-buffering.

The four combinations in the higher-cost-but-service-preserved quadrant share a different profile: the actual baseline was already cost-efficient, typically because the planner had been operating below the analytical reorder point through manual override, and the optimization moves toward but does not cross under the current operational point. In these cases, the DDR is buying back service at a cost premium, in line with its objective of maintaining or improving the training-period in-stock floor. The increase is real but bounded: service rises materially across the group, and the cost premium represents a defensible trade-off rather than a failure of the policy.

The two combinations in the unfavorable quadrant share a single diagnostic: the test period contains a demand pattern that breaks materially from the training distribution, so the reorder point optimized on training delivers neither the buffer needed for the realized demand, nor the cost discipline expected from a tighter setup. These cases are the clearest evidence that Stage 1 alone is not sufficient: a static reorder point, analytical or data-driven, cannot adapt to a regime change inside the test window, which is precisely what motivates the Stage 2 work in Section 4.2.2.

Tier 2 Results

The new fixed forecast horizon of 4 months is applied to the ten Tier 2 MTS combinations. The evaluated policies conditions are the same than described in Tier 1 Results. Table 6 reports portfolio averages across the ten Tier 2 MTS combinations evaluated over the 12-month test window.

Table 6. Tier 2 PPCs portfolio comparison – Stage 1 reorder-point optimization.

KPI	Actual	Forecast-Based (APO)	DDR Policy 1
Re-Order Point (kg)	–	17,449	7,431
Portfolio avg. on-hand (kg)	18,933	25,846	20,012
In-stock (%)	93.4%	98.9%	92.7%
Replenishment orders / yr	5.1	5.0	3.5
Avg. order size (kg)	21,949	23,684	24,222
Est. annual holding cost (€)	6,722	9,148	7,016

Two findings emerge at the portfolio level. The first is the cost of the analytical reorder-point as currently parameterized: applied without modification across the ten combinations, it carries 37% more inventory than the actual baseline and raises annual holding cost from €6,722 to €9,148, a 36% increase. Service improves from 93.4% to 98.9%, a 5.5 percentage-point gain, but it is purchased almost entirely through additional buffer stock rather than through a change in replenishment frequency: replenishment orders per year remain essentially flat (5.1 → 5.0), so the analytical setup absorbs the service uplift in a larger reorder point rather than in a tighter trigger cadence.

The second finding is what the optimization recovers: the data-driven replica reduces the reorder point from 17,449 kg to 7,431 kg (-57%) and brings average on-hand inventory back to within 6% of the actual baseline (20,012 kg vs. 18,933 kg), against an in-stock level of 92.7%: 0.6 percentage points below the actual baseline and 6.2 percentage points below the analytical APO setup. The optimization is bounded so that it cannot drop the reorder point below the σ-floor implied by forecast-error variability, which is what limits the service erosion to less than one percentage point. Relative to the analytical APO setup, the data-driven replica releases 23% of the inventory buffer; relative to the actual baseline, it operates at approximately equal inventory and approximately equal service.

The portfolio average masks heterogeneity across combinations, classified into the same four cost–service quadrants used at Tier 1, with cost measured as average on-hand inventory and service as in-stock percentage relative to the actual baseline. The breakdown is shown in Figure 4.

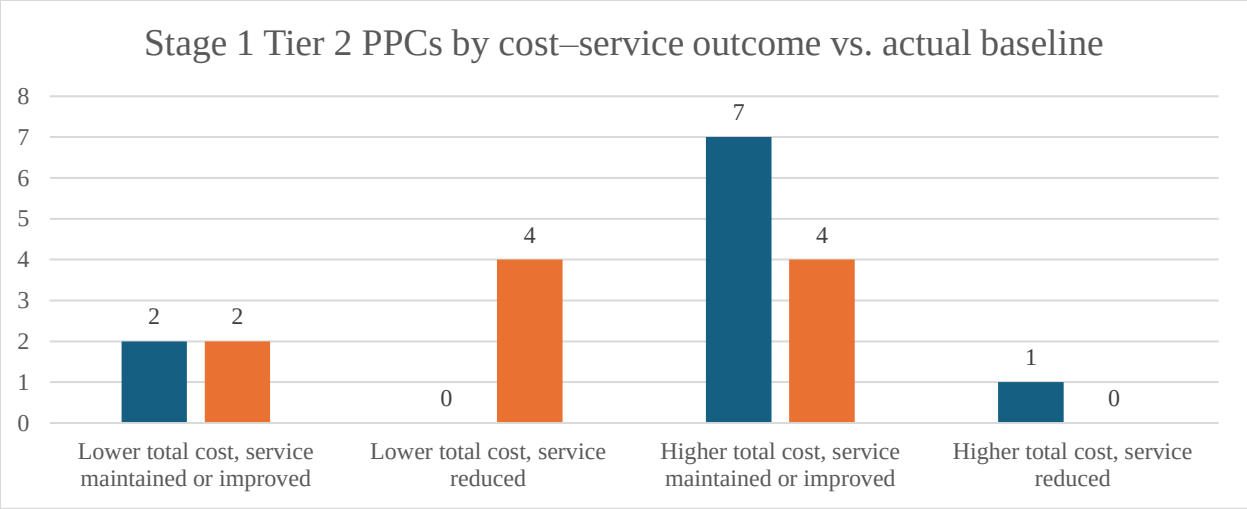


Figure 4. Distribution of the 10 Tier 2 PPCs across the four cost–service quadrants under each policy, benchmarked against the actual on-hand baseline.

The pattern at Tier 2 is the inverse of Tier 1. Under the analytical APO setup, only two of ten combinations land in the unambiguous-improvement quadrant; seven sit in the higher-cost / service-maintained-or-improved quadrant, the analytical reorder point is buying back service at an inventory premium, and one falls into the unfavorable quadrant. The data-driven optimization redistributes these cases: still two combinations in the unambiguous-improvement quadrant, but four move into the lower-cost / service-reduced quadrant as the tighter reorder point releases inventory at the expense of small service erosions, and four remain in the higher-cost / service-maintained quadrant.

No combination lands in the unfavorable quadrant under the data-driven setup. The interpretation is consistent with the portfolio-level result: the analytical APO setup at Tier 2 systematically over-buffers, and the data-driven optimization corrects for this, but it does so by trading marginal service for inventory release rather than by improving both dimensions simultaneously. This is the Tier 2 counterpart to the trade-off surfaced at Tier 1, and it sets up the case for Stage 2, where allowing the policy structure itself to change (Order-Up-To with a tunable review period) is expected to recover service without giving back the inventory release.

4.2.2. Stage 2 – Flexible-Horizon and Cross-Policy Optimization

Stage 2 relaxes the constraints that Stage 1 held fixed: the forecast horizon is no longer locked at the current SAP APO value, and the policy itself is allowed to change. Both tiers' policy candidates are re-evaluated under per-combination parameter optimization, with the forecast horizon swept over $f \in \{1,\ldots,12\}$; at Tier 1, the (s, S) parameters for Policy 2 are derived analytically from training-period demand and the master-data EOQ; at Tier 2, the review period for Policy 2 is swept over $R \in \{1,\ldots,12\}$ months. The aim is to evaluate, on the same MTS combinations as Stage 1 plus the ten Tier 2 combinations, what the combined effect of the tier-specific policy choice and per-combination parameter optimization delivers when neither the planning architecture nor the parameter regime is held to the current SAP APO setup.

Tier 1 Results

At Tier 1, two policies are simulated under identical demand sequences and identical shared parameters (lead time, target service level = 95%, frozen horizon = 1 period). **Policy 1: Forecast-Based Reorder-Point** works as described in Stage 1, with the forecast horizon optimized per combination. **Policy 2: (s, S) with EOQ-bounded order quantity** ignores the forecast at the trigger and orders an analytically derived EOQ-bounded campaign batch whenever inventory position falls below the reorder level s. Both policies are reported side by side; the planner selects the parameter values and policy to deploy per combination.

Table 7. Tier 1 PPCs portfolio comparison – Stage 2 evaluation under optimized parameters.

KPI	Actual	P1 – Forecast-Based	P2 – (s, S)
Portfolio avg. on-hand (kg)	139,829	171,381	111,594
In-stock (%)	97.5%	100.0%	92.0%
Production campaigns / yr	3.07	3.16	3.72
Avg. order size (kg)	187,281	195,098	153,735
Est. annual holding cost (€)	48,365	59,584	38,910
Est. annual changeover cost (€)	71,774	73,898	86,851
Total cost (€)	120,139	133,481	125,761

Table 7 reports portfolio averages KPIs. Policy 1, with per-combination optimized horizons, raises in-stock performance from 97.5% to 100.0%, making every Tier 1 PPC stockout-free over the test window. This service improvement comes at an 11% total-cost premium, driven almost entirely by holding cost. Changeover cost is essentially unchanged, confirming that optimization buys perfect service through a larger buffer rather than through more frequent triggers.

Policy 2 inverts this trade-off: average inventory falls 20% versus the actual baseline and holding cost drops to €38,910, the lowest value in the comparison. However, service decreases to 92.0% and changeover cost rises 21% as the EOQ-bounded campaign batch creates more production cycles during the test window. Neither policy improves on the actual baseline along both the cost and service dimensions at portfolio level. The value of the framework at Tier 1 is therefore not to identify a single dominating policy, but to surface the trade-off transparently per combination.

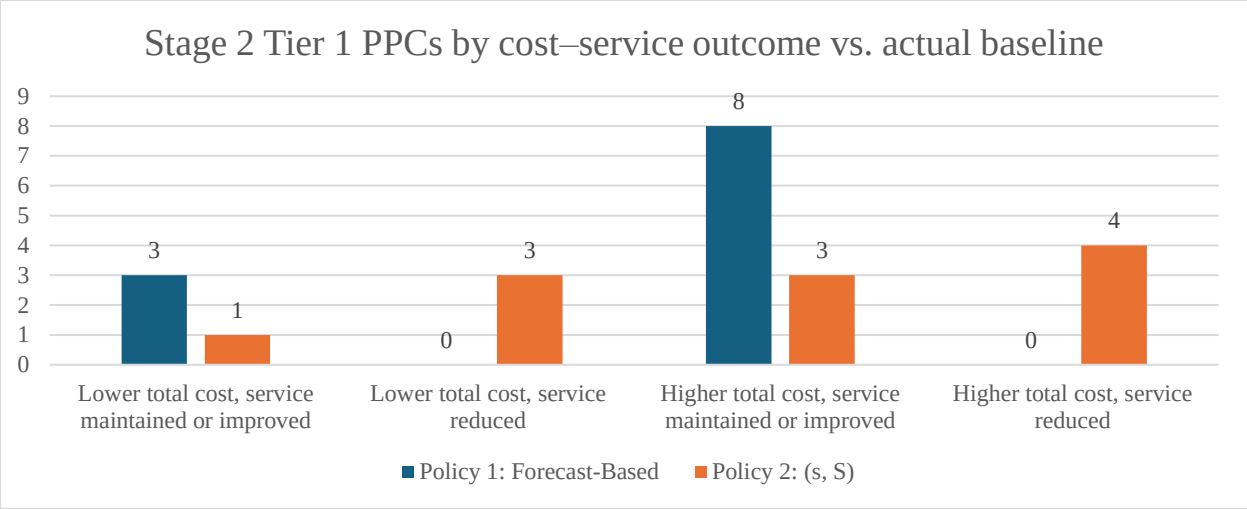


Figure 5. Same 11 Tier 1 PPCs reclassified under Stage 2's relaxed parameters, comparing the per-combination optimized Forecast-Based policy against the (s, S) alternative.

Policy 1's profile is service-first: it preserves or improves service on all eleven combinations, and three combinations also reduce total cost, putting them in the unambiguous-improvement quadrant. The remaining eight cost more than the actual baseline because the analytically derived reorder point sits higher than the parameter values the planner has been operating at, often through manual override. Policy 2's profile is the opposite: cost-first with significant service exposure. Only one combination lands in the unambiguous-improvement quadrant; three reduce cost at the expense of service, three carry higher cost while preserving service, and four sit in the unfavorable quadrant where the policy degrades both dimensions simultaneously.

The unfavorable cases share the diagnostic surfaced in Stage 1: the test period contains demand realizations that depart materially from the training distribution, so the analytically derived (s, S) parameters fit the trained profile but not the realized one. The (s, S) policy is therefore the right choice on combinations with stable demand and high changeover cost, where the EOQ-bounded campaign batch does the work, but should not be deployed without the planner reviewing the training-vs-test demand profile.

Tier 2 Results

The same comparison is applied to the ten Tier 2 MTS combinations. Two policies are evaluated under per-combination optimized parameters: Policy 1 – Forecast-Based with the forecast horizon swept, and Policy 2 – Order-Up-To (S) with the review period swept over $R \in \{1,\ldots,12\}$ months. The recommendation engine assigns Policy 1 to eight combinations and Policy 2 to two; results are reported at portfolio level under the recommended policy per combination.

Table 8. Tier 2 PPCs portfolio comparison – Stage 2 evaluation under optimized parameters.

KPI	Actual	P1 – Forecast-Based	P2 – Order-Up-To (S)
Portfolio avg. on-hand (kg)	15,610	12,952	17,265

KPI	Actual	P1 – Forecast-Based	P2 – Order-Up-To (S)
In-stock (%)	83.2%	90.8%	89.1%
Order frequency (orders/yr)	9.4	7.6	7.2
Avg. order quantity (kg)	8,372	11,746	11,436
Total quantity ordered (kg)	112,500	97,525	72,630
Est. annual holding cost (€)	5,541	4,555	6,117

Unlike Tier 1, the improvement at Tier 2 is unambiguous. Policy 1 reduces average inventory by 17% (15,610 → 12,952 kg), raises in-stock from 83.2% to 90.8% (+7.6 percentage points), and cuts holding cost by 18% (€5,541 → €4,555). All three KPIs move in the favorable direction simultaneously, and Policy 1 is the recommended configuration for eight of the ten combinations. The Tier 2 baseline was operating with low service slack and inflated inventory at the same time, so the optimization has room to work on both dimensions without facing the cost-service trade-off observed at Tier 1.

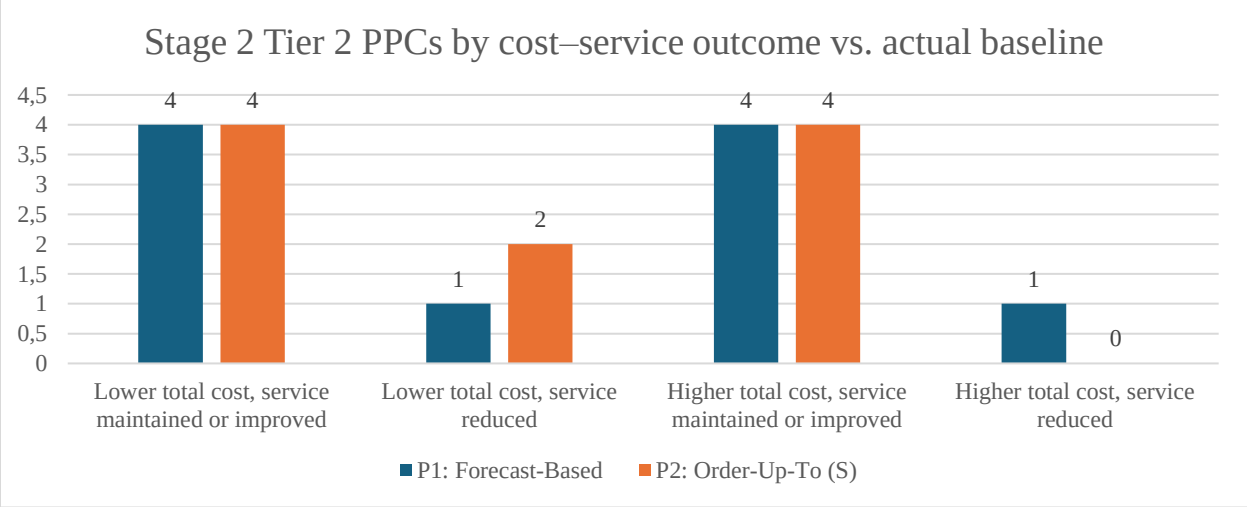


Figure 6 Distribution of the 10 Tier 2 PPCs reclassified under Stage 2's relaxed parameters, comparing the per-combination optimized Forecast-Based policy against the (S) alternative.

Under Policy 1, eight of the ten combinations land in service-favorable quadrants and four of these also reduce inventory, putting them in the unambiguous-improvement quadrant. Only two combinations sit in service-reduced territory, with one carrying both higher cost and reduced service. Policy 2 produces a similar but slightly different service profile: eight service-favorable cases and zero combinations in the unfavorable quadrant, with the trade-off shifted toward two cases of lower-cost-but-reduced-service. Both policies are structurally favorable at Tier 2, and the recommendation engine sorts between them based on demand profile rather than compensating for a structurally weak policy.

Stage 2 surfaces an additional output that is structurally tied to the joint optimization of forecast and frozen horizons: the average container fill rate per combination, computed as the average order quantity divided by the full-container-load volume from the master-data file. Across both tiers, fill rates under the recommended policy span a wide range (from below 30% on intermittent low-volume combinations to above 200% on combinations whose order quantities cover multiple containers) with substantial movement relative to the SAP APO baseline driven by the horizon optimization.

This makes Stage 2 directly useful for an objective the existing parameterization does not expose: identifying combinations whose container utilization is structurally inefficient and resizing the order cadence to bring them into a defensible range. When the optimization scope is fill-rate consolidation rather than cost reduction, the Stage 2 framework produces actionable per-combination recommendations that Stage 1 cannot, since Stage 1 holds the horizon fixed.

The two tiers produce structurally different findings. At Tier 1, the actual baseline operates at 97.5% in-stock, and any cost reduction is paid for in availability; the framework's value is to surface the trade-off transparently rather than to deliver a dominating policy. At Tier 2, the actual baseline operates at 83.2% in-stock with inflated inventory, and the framework reduces inventory, raises service, and cuts holding cost simultaneously. The asymmetry is informative: it tells the planner that automated parameter optimization adds the most value where the current parameterizations furthest from a defensible service-cost frontier. At Tier 1 the planner is already near that frontier; at Tier 2 they are not.

4.3. Decision Support Tool

The contribution of this thesis is not only the analytical framework of Section 4.2 but the planning tool that operationalizes it. Four design principles distinguish the tool from a one-off study and earn the framework its place in the planner's workflow; each is realized in a specific architectural choice. The visual walkthrough of every screen referenced below is documented in Appendix E.

Modularity. The workflow is decomposed into five independent modules (Figure 7) (Import Data, Demand Analysis, Supply Analysis, Simulation, Batch Export), each producing an output the next consumes but executable in isolation for diagnostic purposes. The data layer mirrors this with four loaders, one per SAP extract, and a shared normalization function applied at every join key. The Stage 2 forecast-horizon optimization was added after the Stage 1 implementation was already in production at the sponsor's site by modifying only the simulation module, with no impact on upstream forecasting or downstream KPI dashboard.

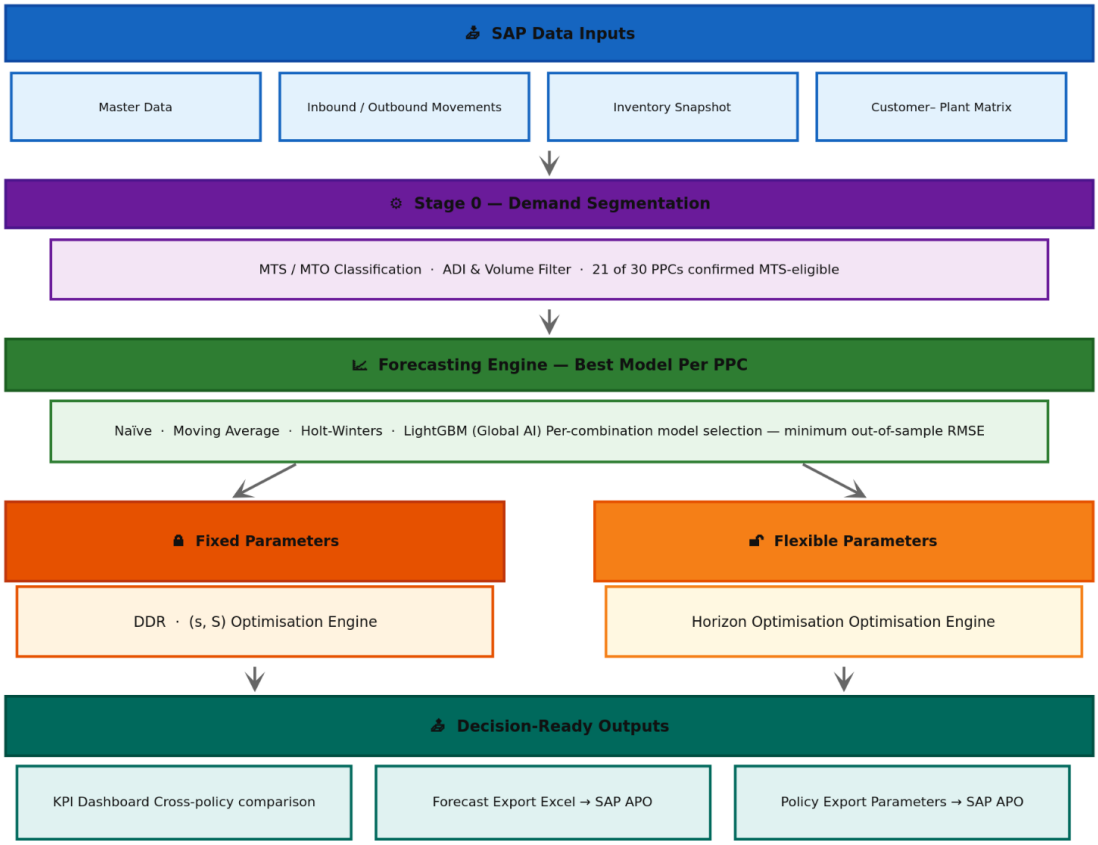


Figure 7. Decision-Support Tool Modules.

Repeatability. Every parameter the simulation reads is derived from data: lead time as the volume-weighted average from historical inbound shipments, safety stock as $z \times \sigma_{LT}$ from forecast-error variability, EOQ from the master-data file, average production interval from EOQ and mean daily demand. The forecast evaluation runs on a fixed train/test split with deterministic seeds. The same SAP extract uploaded a second time produces identical recommendations and identical per-combination quadrant assignments, which is what makes the framework auditable: any recommendation challenged in a replenishment review can be traced to a specific data record.

Override-friendly defaults. Every parameter is pre-populated with a data-derived default and exposed as an editable control next to its rationale, with the simulation re-running in real time on any change. The MTS/MTO classification surfaces the computed values driving the recommendation ("ADI 1.03 $\leq$ 5, total demand 190,530 kg $\geq$ 50,000 kg $\rightarrow$ MTS") rather than an opaque label, and every cell of the cross-policy comparison can be drilled into at the per-combination level. This positioning, the planner as the decision authority and the tool as the analytical layer that surfaces options, follows the decision-support design philosophy of Turban et al. (2011).

S&OP-cycle execution time. The full pipeline: four-file ingestion, MTS/MTO classification, forecast evaluation, both-policy simulation with parameter optimization, and cross-policy KPI dashboard; completes in under one working hour per product family on a standard laptop, with no manual intervention between stages. The planner uploads the latest SAP extracts at the start of the S&OP cycle, reviews the recommendations as they populate, applies overrides where local context

warrants, and exports the updated parameter workbook before the cycle's planning meeting. Each stage flows directly into the next (Figure 8): forecast outputs feed the simulation, simulation outputs feed the dashboard, and dashboard selections feed the export, with intermediate results cached so adjustments to a downstream parameter do not retrigger the upstream pipeline. This continuity is what makes the framework operationally feasible at S&OP cadence and positions the tool as additive to the existing SAP APO architecture rather than as a replacement for it.

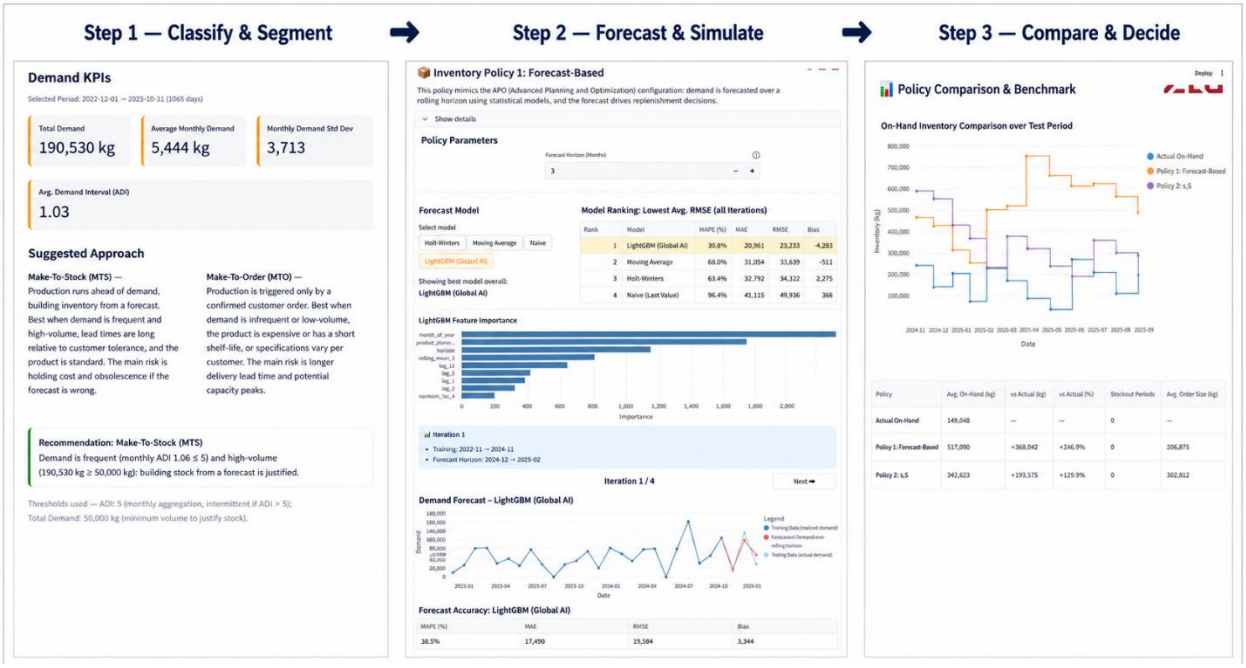


Figure 8. Decision-Support Tool flow visualization.

Conclusion and Recommendations

This thesis addressed the inventory replenishment process of a specialty chemicals multinational that distributes a single high-value product family from one factory to regional warehouses worldwide. The core problem was the absence of a feedback loop between replenishment outcomes and parameter design: safety stock levels and reorder rules were set once, applied uniformly, and never systematically evaluated against alternatives. The response is a Python-based decision-support tool that benchmarks competing replenishment policies against the current SAP APO baseline, integrates a four-method forecasting evaluation (Naive, Moving Average, Holt–Winters, and a state-of-the-art global LightGBM model) directly into the policy simulation, and surfaces the trade-offs to planners as interpretable KPI dashboards.

The framework's contribution is tier-dependent, and the asymmetry is itself the headline finding. At Tier 1, where the actual baseline already runs at 97.5% in-stock near the achievable service-cost frontier, the framework cannot dominate the planner's manual corrective work on both cost and service; its value is to encode that corrective work into a repeatable, auditable process and to make the residual trade-off explicit per combination. At Tier 2, where the actual baseline operates at 83.2% in-stock with inflated inventory, the framework delivers unambiguous joint improvement: −17% inventory, +7.6 percentage points service, and −18% holding cost simultaneously. This asymmetry suggests that automated parameter optimization creates the greatest value where the existing configuration is furthest from a defensible service-cost frontier.

The thesis contributes both an analytical demonstration that a forecasting-inventory pipeline can be operationalized on three years of SAP data without probabilistic forecast distributions or specialized infrastructure, and a production-ready prototype that generalizes to any SKU portfolio for which equivalent SAP extracts are available. Four recommendations follow for the sponsor company, ordered by deployment proximity.

1. **Adopt the Stage 1 reorder-point optimization at Tier 1** as a repeatable substitute for the current ad-hoc override workflow: it reproduces the planner's manual corrective work automatically and produces an auditable trail.
2. **Deploy Stage 2 at Tier 2** as the default replenishment configuration for the ten Tier 2 combinations evaluated, since the framework dominates the actual baseline on all three KPIs simultaneously.
3. **Use the Tier 1 cost-service quadrant view as decision support in S&OP,** not as an automated recommendation: the framework surfaces the trade-off but the side to occupy is the planner's call.
4. **Refresh parameters every S&OP cycle** rather than statically: the optimization runs in under one working hour per product family on the latest SAP extract.

Three limitations bound the conclusions: the simulation replays historical demand rather than sampling from a fitted stochastic model, so the KPI estimates are conditional on the realizations observed during the test window; the analysis covers a single product family over three years, and generalization to other portfolios requires validation of comparable demand characteristics and data quality; and supply-side constraints (campaign minima, vendor fill rates, transport capacity) are not represented in the simulation engine. Two near-term extensions follow naturally from the existing infrastructure: adding a Croston/SBA forecaster to extend the framework to make-to-order

combinations and running the Tier 2 simulation across all downstream plants simultaneously to capture network-level interdependencies. A separate research direction, beyond the deployment scope of this thesis, is to investigate whether the predict-then-optimize architecture can eventually be replaced by an end-to-end deep reinforcement learning policy seeded on the same SAP data and the same simulation infrastructure developed here. An approach for which Qi et al. (2023), Madeka et al. (2022), and Boute et al. (2022) have established both the feasibility and the design roadmap. The thesis therefore does not represent a closed project but the first stage of a data-driven inventory optimization capability that can evolve from structured policy benchmarking toward learning-based replenishment as data accumulates and organizational confidence in the framework grows.

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Appendices

Appendix A: Project Management

The project followed an Agile methodology with the core simulation engine built from the earliest stages, so that planners and managers at the sponsor company could interact with a functional tool throughout development. The continuous feedback loop shaped the policy logic, the KPI definitions, and the planner workflow described in Chapter 4 directly in response to the operational reality at the sponsor, rather than in response to a fixed specification written upfront.

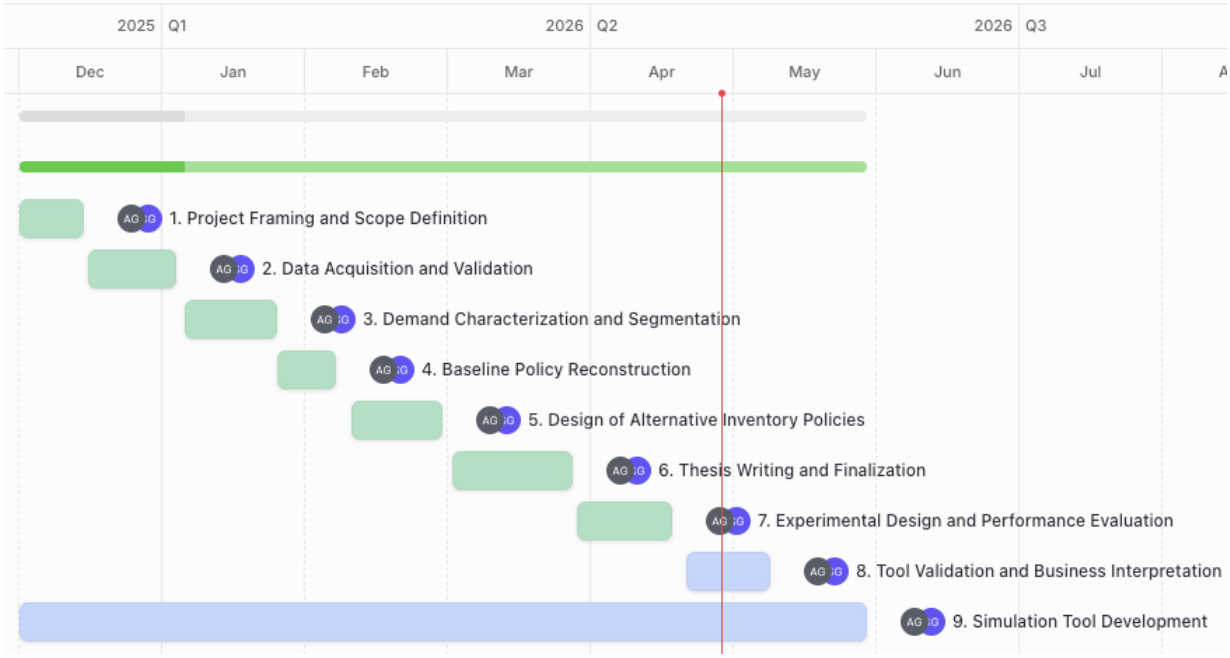


Figure 9. Thesis project Gantt diagram.

Appendix B: Data and Preprocessing

The empirical analysis draws on four operational datasets extracted directly from the sponsor company's SAP system, covering October 2022 to September 2025. Together they provide a complete picture of the physical flow of material through the network: what was ordered and received, what was shipped to end customers, what stock was held at each location, and which customers are served by which plants. Table 9 lists the four files and the columns consumed downstream.

Table 9. Source datasets required by the tool.

File: Sheet(s)	Key columns	Purpose
Master-data: Single sheet	Material, Material Name, Product, Pallet Size, Full Container Load, Actual optimal campaign size, Set-up	SKU attributes used for cost calculations (changeover, holding), lot-size constraints, and unit cost allocation across batches.

File: Sheet(s)	Key columns	Purpose
	Costs, Holding Costs, Unit Costs.	
Movements Data: *Regional_Wareho_101* (receipts) and *Customer_issues_601* (outbound shipments)	Material, Plant, Calendar Date, Receipt/Issue Total Stock, batch, Movement Type, customer, Country	Core transactional dataset. Movement type 101 records batch receipts at each plant; type 601 records end-customer outbound shipments. All demand analysis, inventory simulation, and LightGBM training operate exclusively on type-601 records.
Inventory: Single sheet	0MATERIAL, 0PLANT, 0CALMONTH, Quantity of stock on hand	Opening stock-on-hand positions per material and plant at the reference period start. Used to initialize inventory trajectories and as a reference line in the Supply Analysis and Simulation views.
Customer–Plant Matrix: Single sheet	Customer-to-plant mapping (no fixed column schema enforced)	Maps end-customers to their supplying plants. Used in combination with the Movements Data to classify direct-shipment records at the production site.

Two preprocessing rules deserve mention because they shape every downstream number in this thesis. First, material and plant codes appear in inconsistent formats across the SAP extracts which would otherwise cause silent join failures between the four files. A shared normalization function is applied at every join key: the input is converted to string, whitespace is stripped, and a trailing .0 is removed. Second, production-site records in the movements file mix end-customer shipments with inter-plant warehouse-replenishment transfers under the same movement type 601. These are separated by a customer-code-length filter specific to the sponsor's SAP configuration: at production sites, only records with a customer code longer than four characters are treated as genuine end-customer demand; shorter codes identify warehouse destinations and are excluded from the demand analysis. Without this filter, demand at production sites would be roughly double the true end-customer figure.

Beyond the four raw loaders, the data layer derives two computed series consumed by the simulation. The inventory trajectory function reconstructs the daily on-hand position for any product–plant combination by replaying all 101 and 601 movements against the opening stock from the inventory snapshot; the resulting daily series is the actual on-hand reference line in the simulation charts.

Appendix C: LightGBM Global Model – Training Detail

Section 4.1 describes the two design choices that drive accuracy on the sponsor's portfolio (the direct multi-step strategy and the intermittency-aware feature set). This appendix documents the remaining elements of the LightGBM specification: the eligibility filter, the training procedure including walk-forward selection of the boosting rounds and inverse-volume sample weighting, and the prediction-time fallback logic. Together with Section 4.1 it constitutes the complete model description.

Eligibility filter and tail alignment: A combination is excluded from training if it has fewer than 12 months of total history or fewer than 6 non-zero demand months, since very short or near-entirely-zero series produce almost no valid lag samples and add noise without learnable signal. Excluded combinations receive the mean-fallback prediction described below. Before feature engineering, every combination's monthly series is extended to the global last month of the dataset by filling trailing periods with zero, so that products that stopped receiving orders before the end of the observation window still contribute complete-length sequences and the model sees zero-demand tails as informative signal rather than missing observations.

Walk-forward selection of n_estimators: The number of boosting rounds is the most consequential hyperparameter, and the optimal value is determined per session through 3-fold walk-forward cross-validation. In fold k, the model trains on the first 60% + k·10% of the time range and validates on the next 10% window, with early stopping at patience 50 identifying the best iteration per fold. The mean best iteration across folds is scaled by 1.10 to account for the final model training on more data than any single fold and clipped to [50, 1000]. If the dataset has fewer than 10 distinct prediction dates, the procedure falls back to 300 rounds.

Inverse-volume sample weights: Raw training data is dominated by high-volume combinations: large absolute demand values produce proportionally larger squared errors, so a standard RMSE objective would implicitly down-weight smaller combinations. Each training sample is therefore weighted by the inverse of its combination's mean individual order size (computed from raw 601 movement lines, not monthly aggregates), normalized so that the mean weight across all samples equals 1. This keeps the effective learning rate unchanged while equalizing the loss contribution of small and large combinations.

Fixed hyperparameters: Learning rate 0.05, num_leaves 63 (a moderately complex tree structure appropriate for this feature set and data volume), min_child_samples 10 (prevents overfitting to very sparse leaf nodes), random_state 42 for reproducibility.

Prediction and fallback logic: At prediction time the full feature vector is computed from the last known observations once, then the model is queried for every horizon $h \in \{1, \ldots, n_periods\}$. Lag values, the rolling mean, and the intermittency fraction are extracted from the actual historical series and held fixed across horizons; only month_of_year and horizon vary. All predictions are clipped at zero. If the product–plant identifier is not found in the label encoder's known classes, because the combination was excluded by the eligibility filter or is genuinely new, the model returns a flat forecast equal to the historical mean of the provided series. The same fallback is applied when the rolling unit is set to weekly, since the model is trained exclusively on monthly aggregates. In both cases the label "LightGBM (Global AI) – mean fallback" is displayed so the planner is aware that the AI model was not used for that particular run.

The full feature set used at every training row and prediction step is summarized in the next table.

Table 10. LightGBM feature set.

Feature	Definition	Rationale
lag_1	Demand at time t	Most recent observed demand; the sole input to the Naive baseline; captures the current demand state
lag_2	Demand at t−1	Allows the model to distinguish a temporary spike (lag_1 high, lag_2 low) from a sustained shift (both high)
lag_3	Demand at t−2	Short-run trend information; together with lag_1 and lag_2, spans the 3-period window of the Moving Average baseline
lag_12	Demand at t−11	Same-month-last-year value; captures annual seasonality without requiring a full seasonal state vector
rolling_mean_3	Mean of demand at t−2, t−1, t	Smoothed level estimate; reduces the influence of a single outlier observation on the short-run signal
nonzero_frac_6	Fraction of the 6 months up to t with demand > 0	Encodes recent intermittency; allows the model to learn that a combination with value 0.33 is likely to produce a near-zero forecast even when lag_1 was non-zero
month_of_year	Calendar month of the target date (1–12)	Captures within-year seasonality without computing explicit seasonal indices
horizon	Steps ahead being predicted (1–24)	The defining feature of the direct strategy; allows the model to learn that forecast uncertainty increases with h
product_plant_encoded	Label-encoded integer for each product-plant combination	Allows the model to learn combination-specific demand levels while sharing its global parameter structure

Appendix E: User Interface Walkthrough

This appendix walks the reader through the decision-support tool in the order a planner uses it: opening the application, importing data and designating Tier 1 plants, inspecting demand, running a single-combination simulation, and finally producing a portfolio-wide batch export. Section 4.1 of the main text describes how the planner reaches results in three steps; this appendix documents what each step looks like on screen.

E.1. Home and Navigation

On launch, the tool presents a landing page with the project title, a one-paragraph summary of the workflow, and a collapsible left-hand sidebar listing the four top-level destinations: Import Data, Demand Analysis, Supply Analysis, Plant Analysis, and Simulation. The Simulation entry expands into two sub-pages: the single-combination simulation cockpit and the Batch Export view. A data-readiness gate disables every analytical destination until the four required SAP files have been uploaded and validated, so the planner cannot reach a chart that depends on data that is not yet loaded.



Figure 10. Landing page and sidebar navigation.

E.2. Data Import and Tier 1 Plant Selection

The Import Data page presents four upload cards in a two-column layout, one per required SAP extract: Master-data, Movements, Inventory snapshot, and Customer–Plant matrix. Each card displays the expected filename and a color-coded border indicating status: grey for empty, amber for uploaded but failing structural validation, green for validated. A Verify Files (Figure 11) button appears only when all four files are present and triggers a two-pass check: a schema check that confirms every required column is present, and a cross-slot check that catches the common error of dropping a file into the wrong slot (e.g. uploading the Master-data file into the Movements slot).

Once all four files are validated, the production-site selector appears below the upload grid (Figure 12). This is the single most consequential configuration step in the tool: the planner designates

which plant codes act as Tier 1 manufacturing nodes (production sites) versus Tier 2 distribution warehouses. This designation drives the tier routing throughout the rest of the tool. The customer-code-length filter at production sites, the choice between (s, S) and Order-Up-To as Policy 2, the lead-time initialization (production lead time at Tier 1, inbound transit time at Tier 2), and the reporting split in the cross-policy comparison tables. Without the designation, downstream analytical pages remain locked.

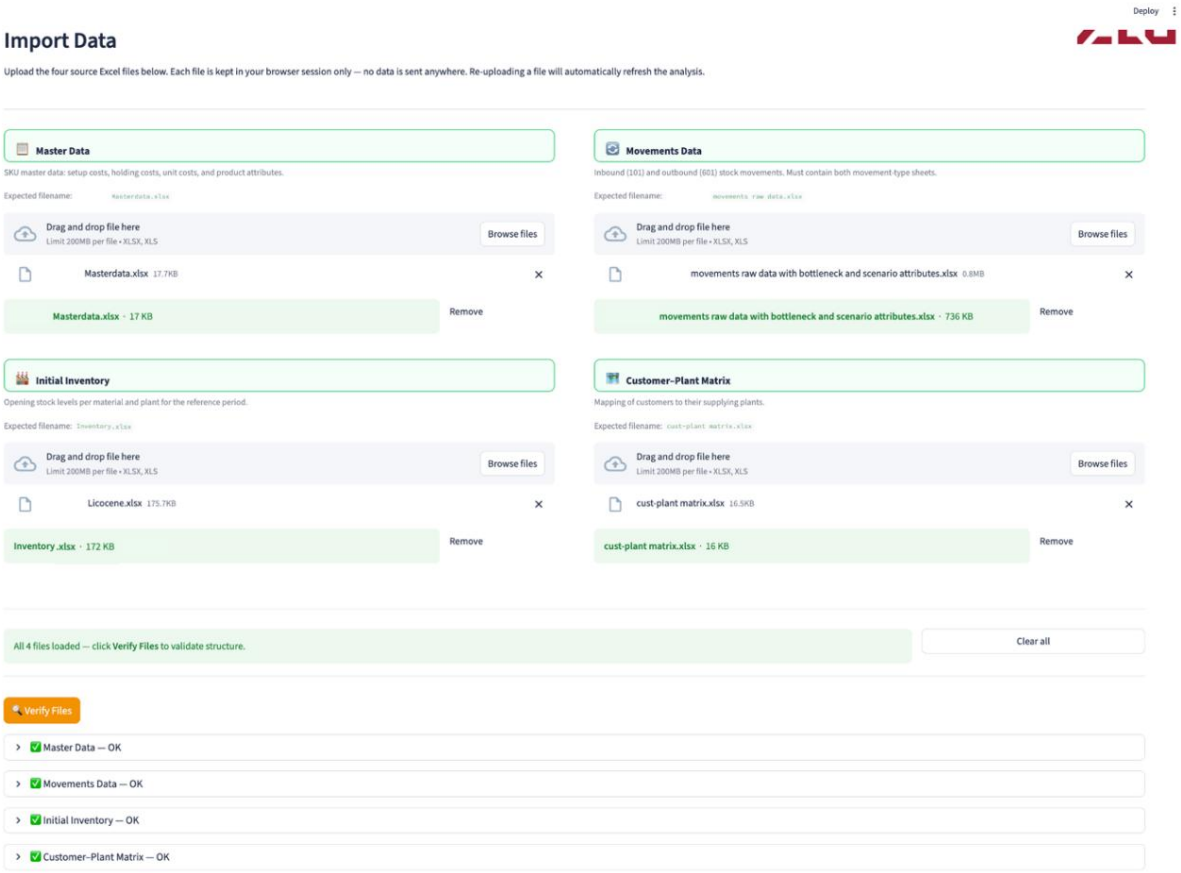


Figure 11. Import Data page: four upload cards with validation status indicators.

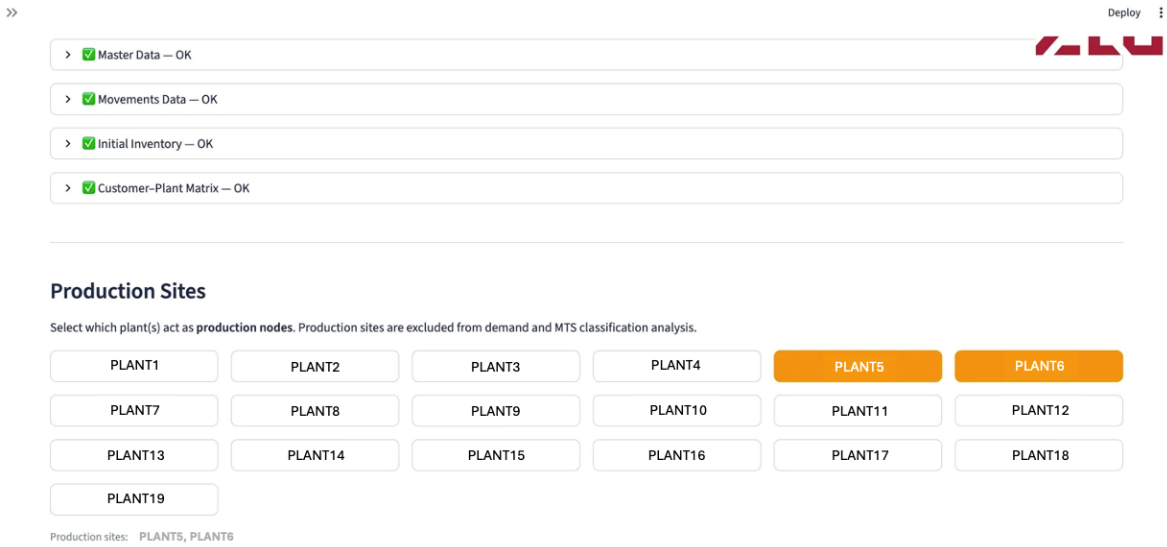


Figure 12. Tier 1 production-site selector.

E.3. Demand and Supply Analysis

Before running any simulation, the planner inspects the Demand Analysis page (Figure 13) to verify that the data behaves as expected: that the volumes match operational intuition, that no plant is missing, and that no anomalous spike or gap requires investigation before influencing downstream forecasting. The page is structured as a left-hand selection panel (SKU dropdown, demand-scope toggle, time-frame selector, date range) and a main panel showing the demand evolution chart with four KPI cards beneath it (Total Demand, Average per period, Standard Deviation, ADI). The demand-scope toggle switches between aggregated global demand for the SKU and demand at a single material–plant combination.

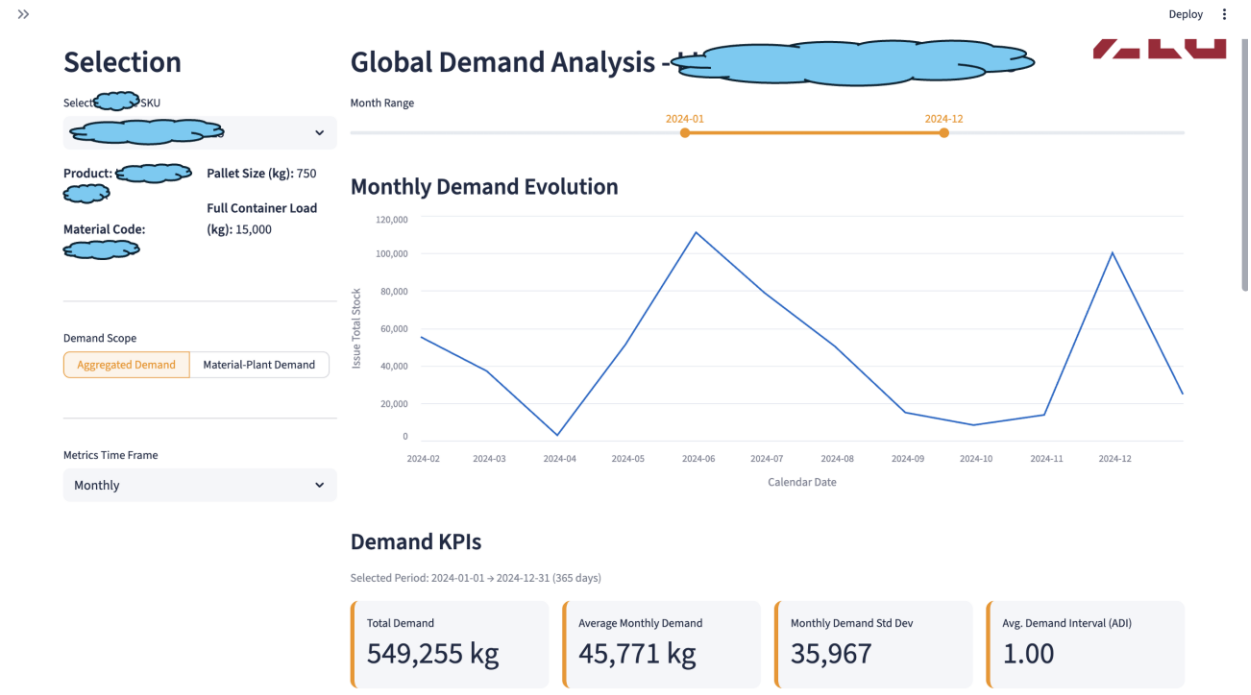


Figure 13. Demand Analysis page.

When the planner is in Material–Plant mode, an MTS/MTO recommendation panel appears (Figure 14) below the KPIs. The panel displays the actual computed values driving the classification (e.g. "ADI 1.03 ≤ 5, total demand 190,530 kg ≥ 50,000 kg → MTS") so the planner sees exactly why the recommendation was reached, not an opaque label. Only combinations classified as MTS are eligible for selection in the Simulation page.

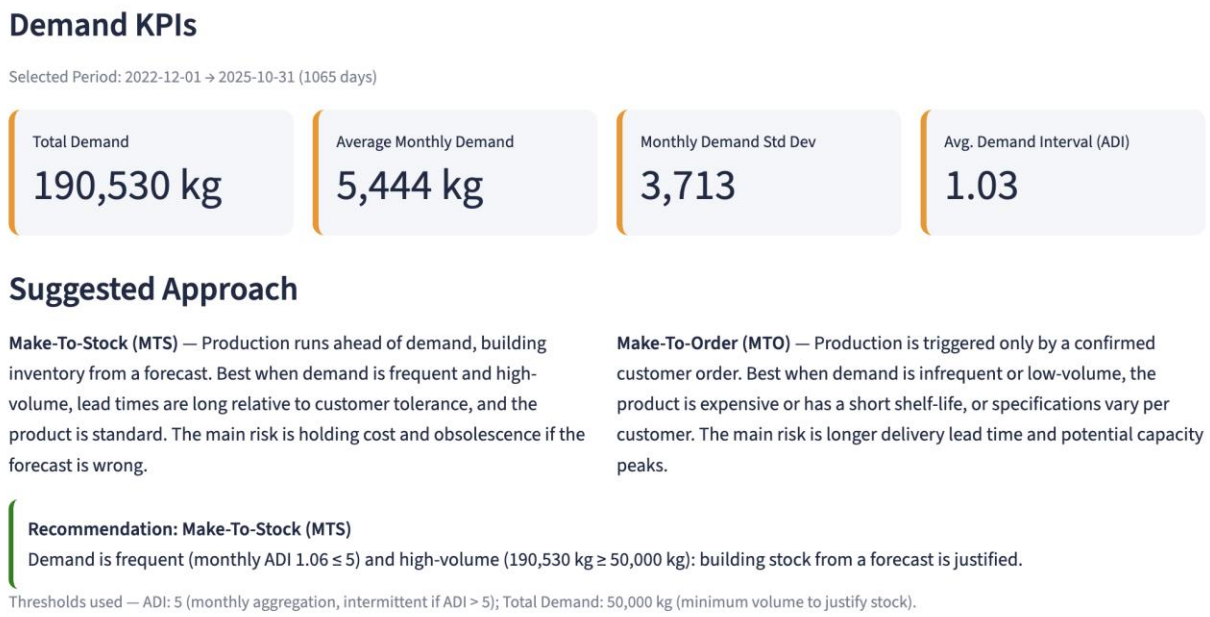


Figure 14. MTS/MTO classification recommendation panel.

When the planner is in aggregated demand mode (no specific plant selected), two pie charts appear below the KPIs (Figure 15): demand by destination country and demand by supplying plant, with production sites annotated to distinguish direct factory-to-customer shipments from warehouse replenishment flows. These views answer whether demand for a SKU is concentrated at a single warehouse or distributed across many: an input to whether a single-combination simulation will be representative.

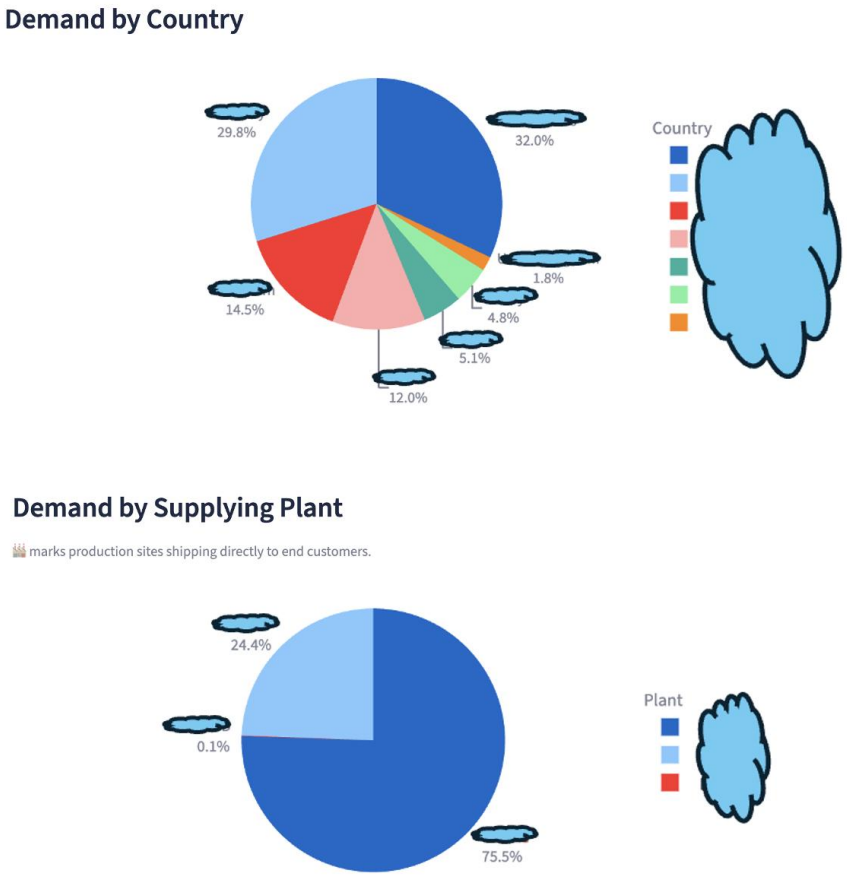


Figure 15. Aggregated demand: country and plant distribution.

The Supply Analysis page (Figure 16) mirrors the demand view from the inbound side. It shows the on-hand inventory trajectory for any combination, the batch receipt table, and the data-derived lead-time estimate: either the EOQ-implied production interval at Tier 1 or the volume-weighted average inbound transit time at Tier 2. The lead time shown here is the same value pre-populated as the default in the simulation cockpit; the Supply Analysis page is the planner's opportunity to validate it before it influences policy parameters.

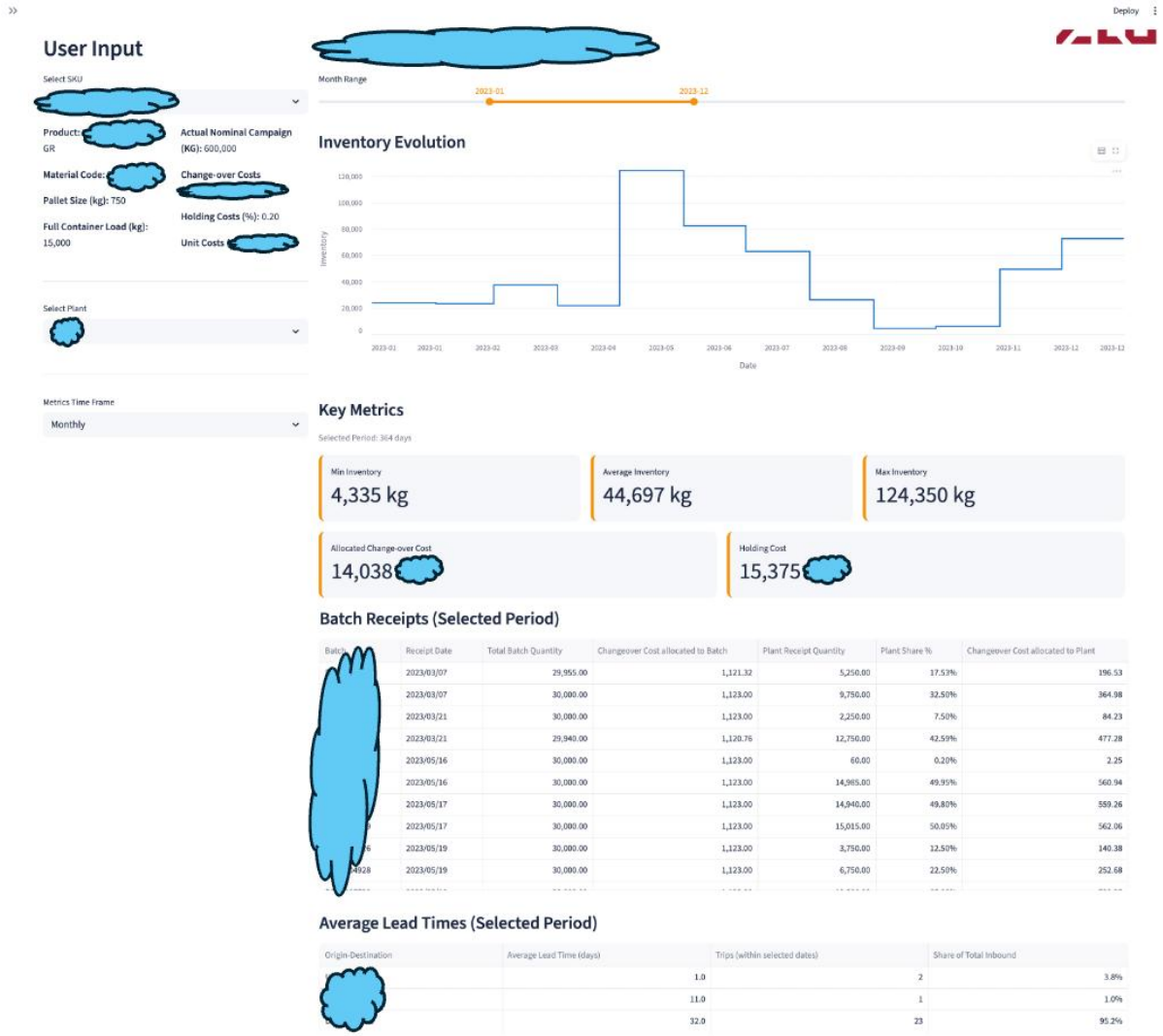


Figure 16. Supply Analysis page: inventory trajectory and lead-time validation.

E.4. Simulation

The Simulation page is where the core analysis runs. It is structured as a multi-step cockpit: the planner first selects an MTS combination, then reviews and (optionally) overrides the shared parameters that apply across all policies for that combination: lead time (days), target service level (slider, 80–99%, default 95%), and production frozen horizon (periods). Every parameter is pre-populated with a data-derived default and the rationale is surfaced inline next to the control.

Below the shared parameters, the planner runs the policy simulations. Each policy renders its own panel showing the simulated on-hand trajectory against the actual SAP-derived baseline, the optimized parameter values selected by the automated grid search (forecast horizon for Policy 1; review period R for the Tier 2 Order-Up-To policy), and the per-policy KPI block. Every optimized parameter is editable and the simulation re-runs in real time on any change, so the planner can interrogate sensitivity by adjusting service level or lead time and watching the trajectory respond.

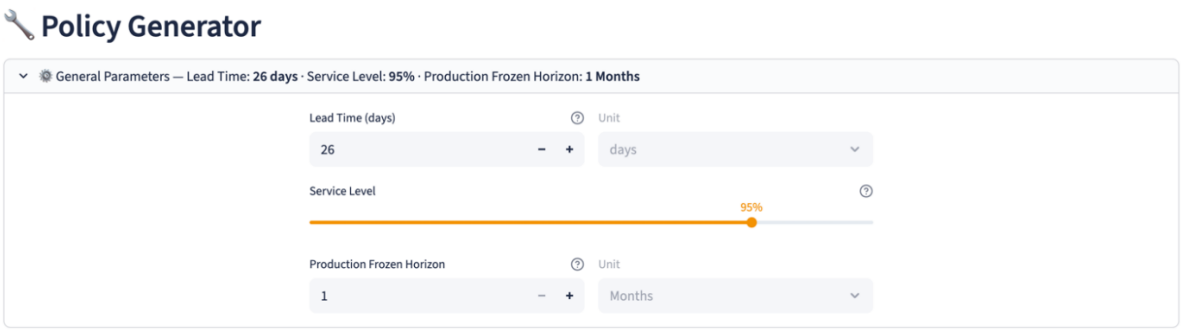


Figure 17. Shared parameters panel: lead time, service level, and production frozen horizon.

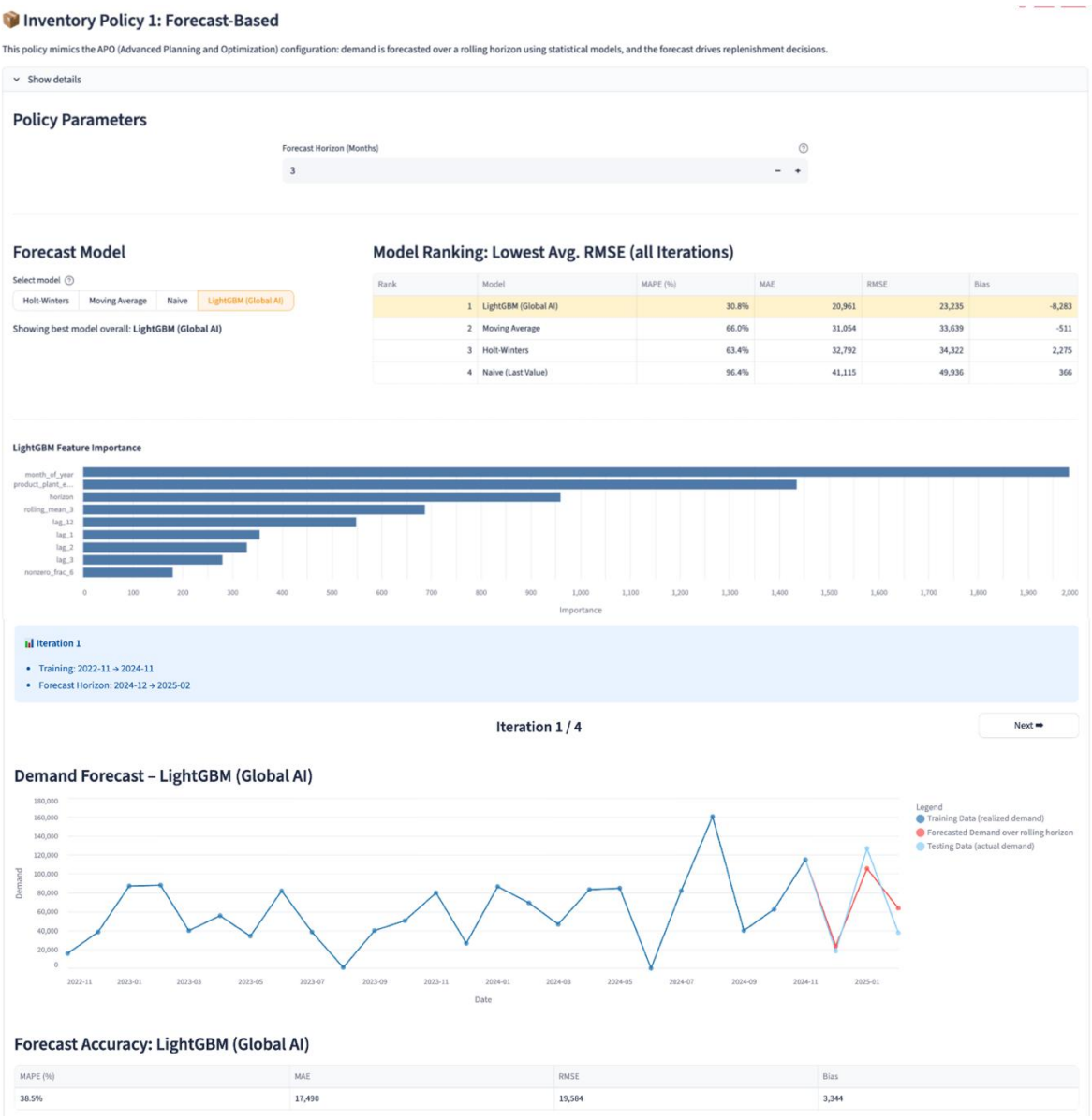


Figure 18. SAP APO Inventory Policy (Tier 1 & 2): Rolling horizon.

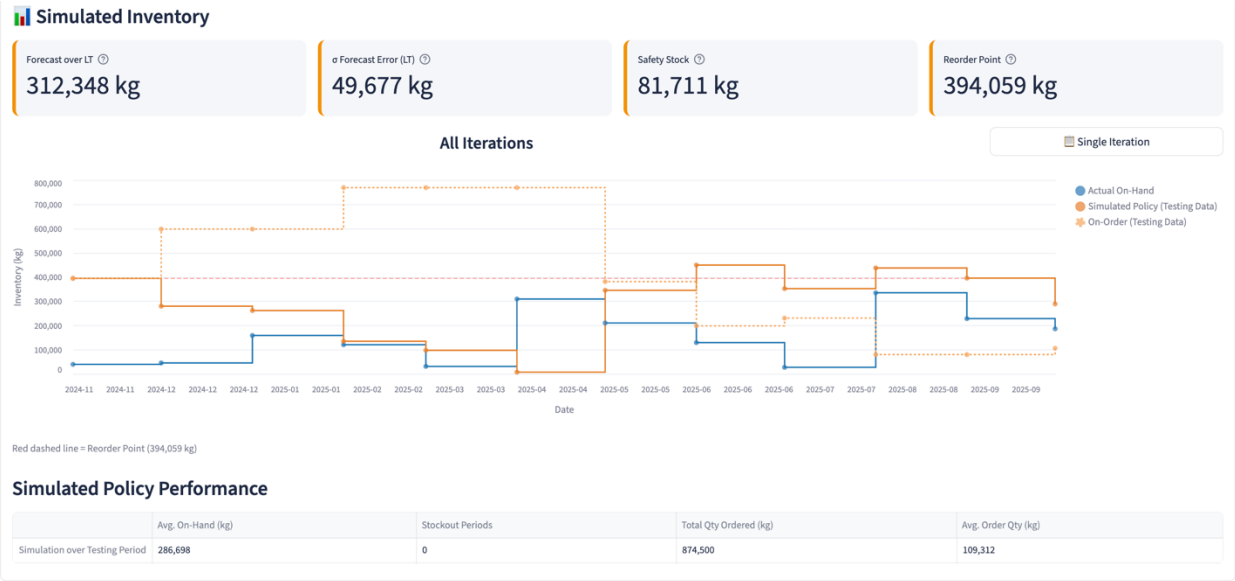


Figure 19. SAP APO Inventory Policy (Tier 1 & 2): Simulated inventory.



Figure 20. (s, S) Inventory Policy (Tier 1): Simulated Inventory.

Once both policies have run, the cross-policy comparison block at the bottom of the page renders the seven-KPI side-by-side table introduced in Section 4.2 (average on-hand, stockout periods, total quantity ordered, average order quantity, order frequency, estimated annual holding cost, estimated annual changeover cost) for the simulated policies and the actual on-hand baseline. This is the screen the planner takes to a replenishment review meeting.

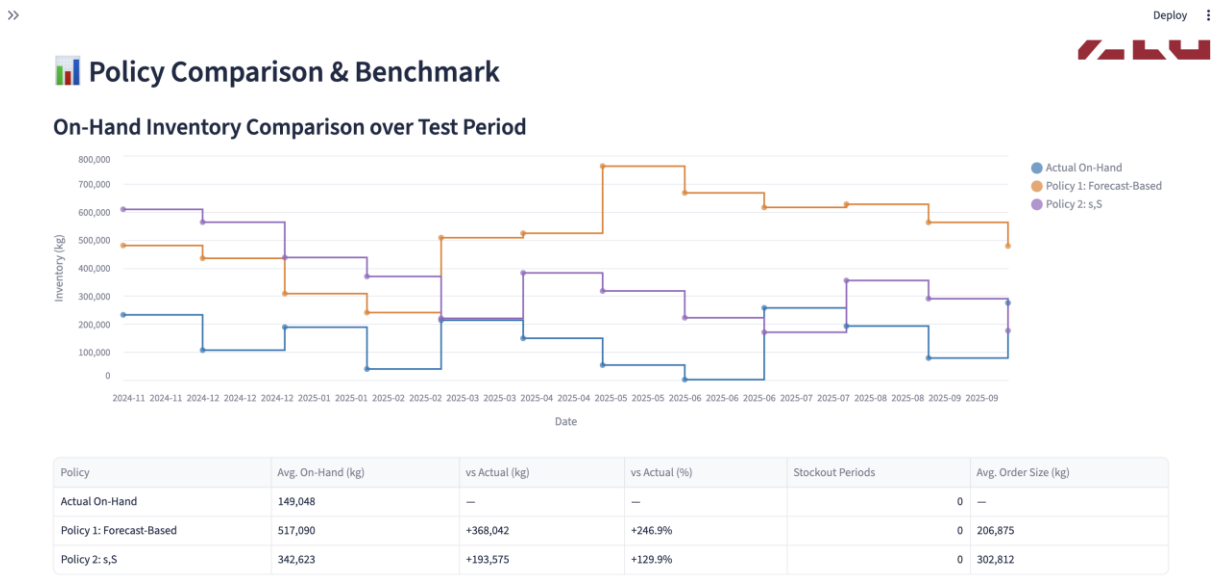


Figure 21. Cross-policy comparison.

E.5. Batch Export

The Batch Export pages scale the same analytics from a single combination to the full MTS portfolio in a single run, producing structured Excel workbooks for use in S&OP planning meetings. The Forecast Export page runs all four forecasting methods across every MTS combination (or every material at aggregated level), evaluates each on a configurable train/test split, and writes one sheet per combination containing the demand history, the forecast series, the four-accuracy metrics (MAPE, MAE, RMSE, Bias), and an embedded chart. The planner selects the forecasting model, the training cutoff, and the number of forward periods to project; no full simulation run is required.

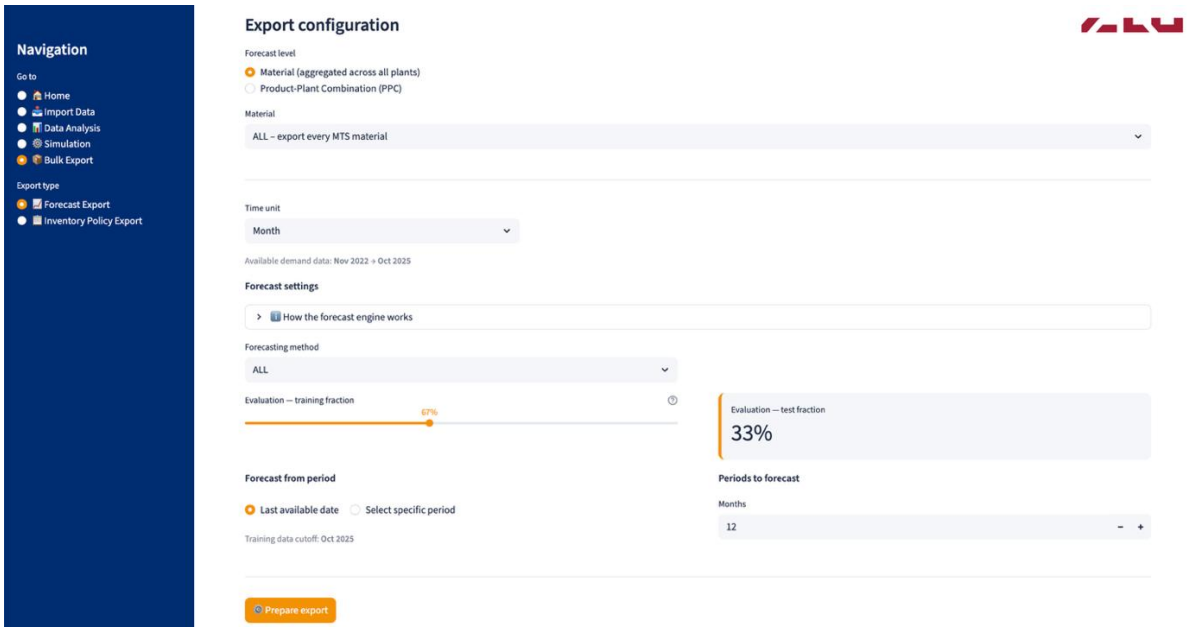


Figure 22. Forecast Export: page configuration.

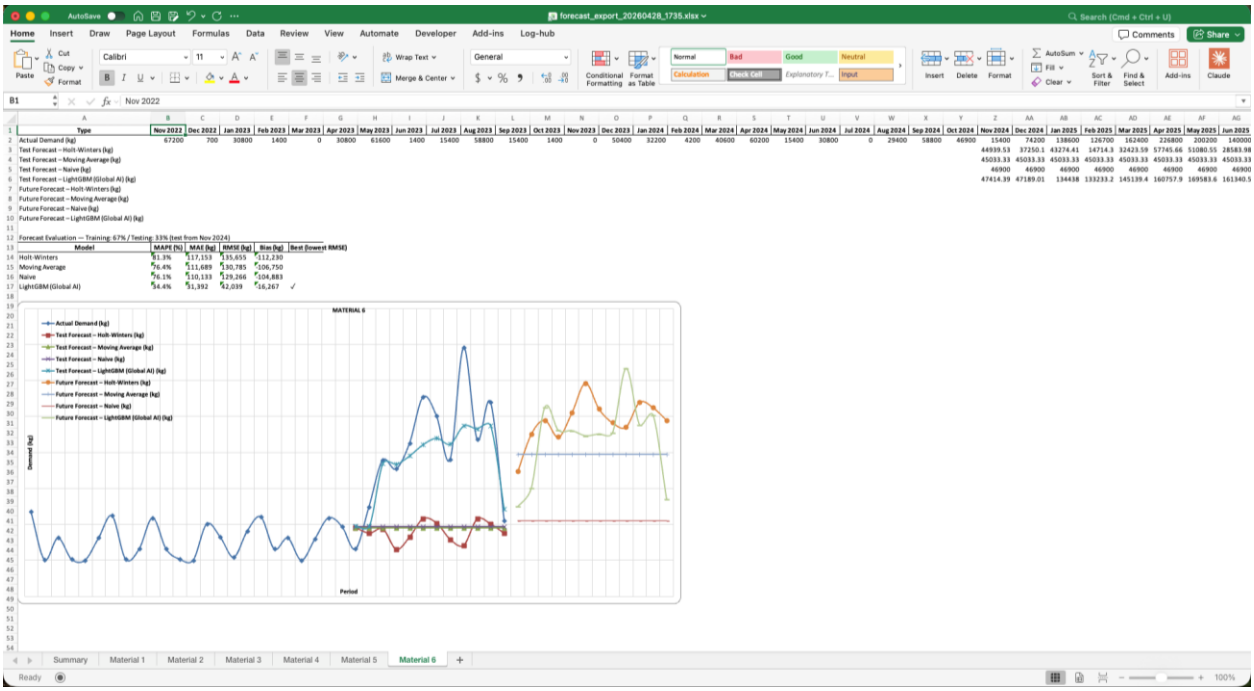


Figure 23. Forecast Export: sample workbook output.

The Inventory Policy Export page computes optimized replenishment parameters for every MTS combination across all policies. Two parameter modes are offered: in auto mode, the tool derives every parameter from training-period demand using the optimization logic of Section 4.2 (forecast horizon for Policy 1 swept over {1, …, 12}; review period R for Tier 2 Order-Up-To swept over {1, …, 12} months); in manual mode, the planner fixes the key parameter for each policy and the tool applies it uniformly across the portfolio. Shared inputs (lead time, service level, frozen horizon) can be set from data or overridden in either mode.

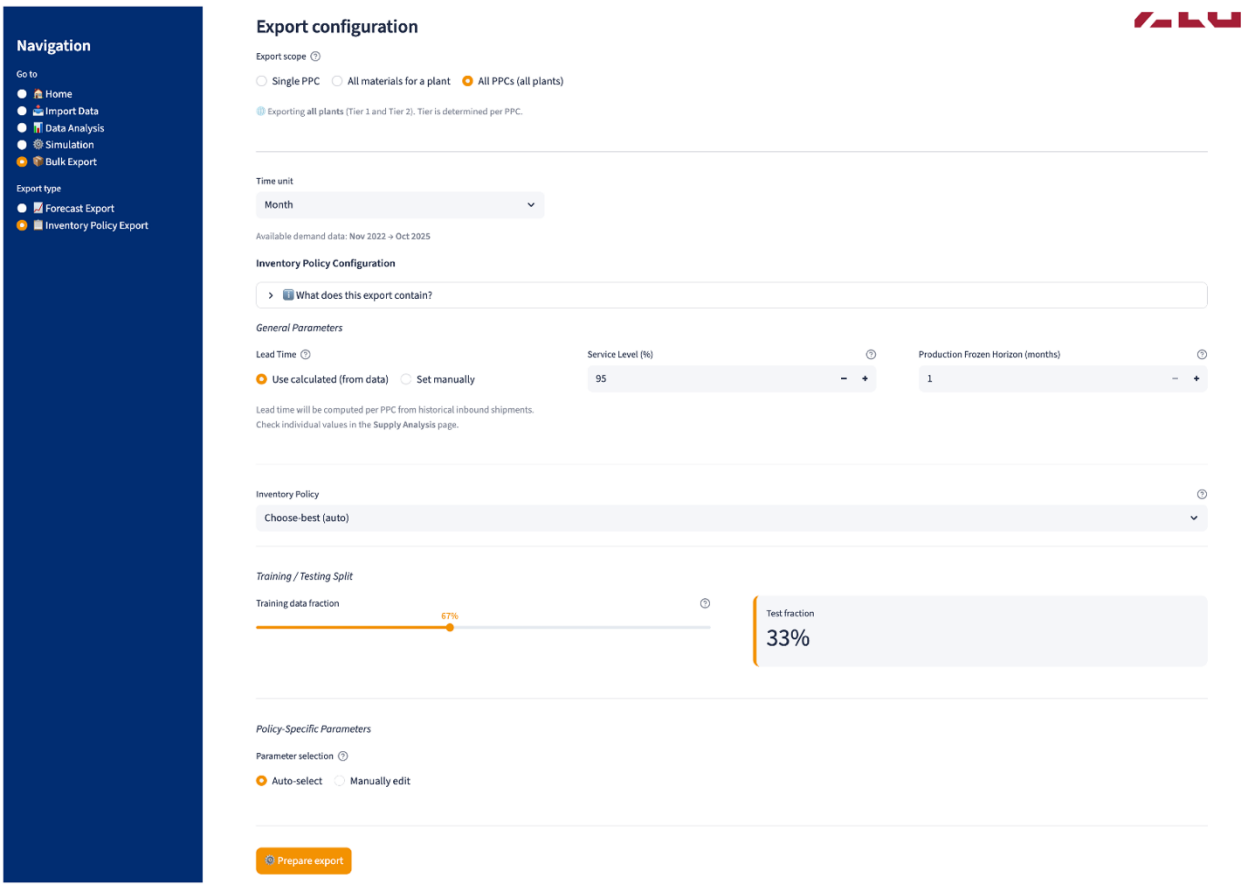
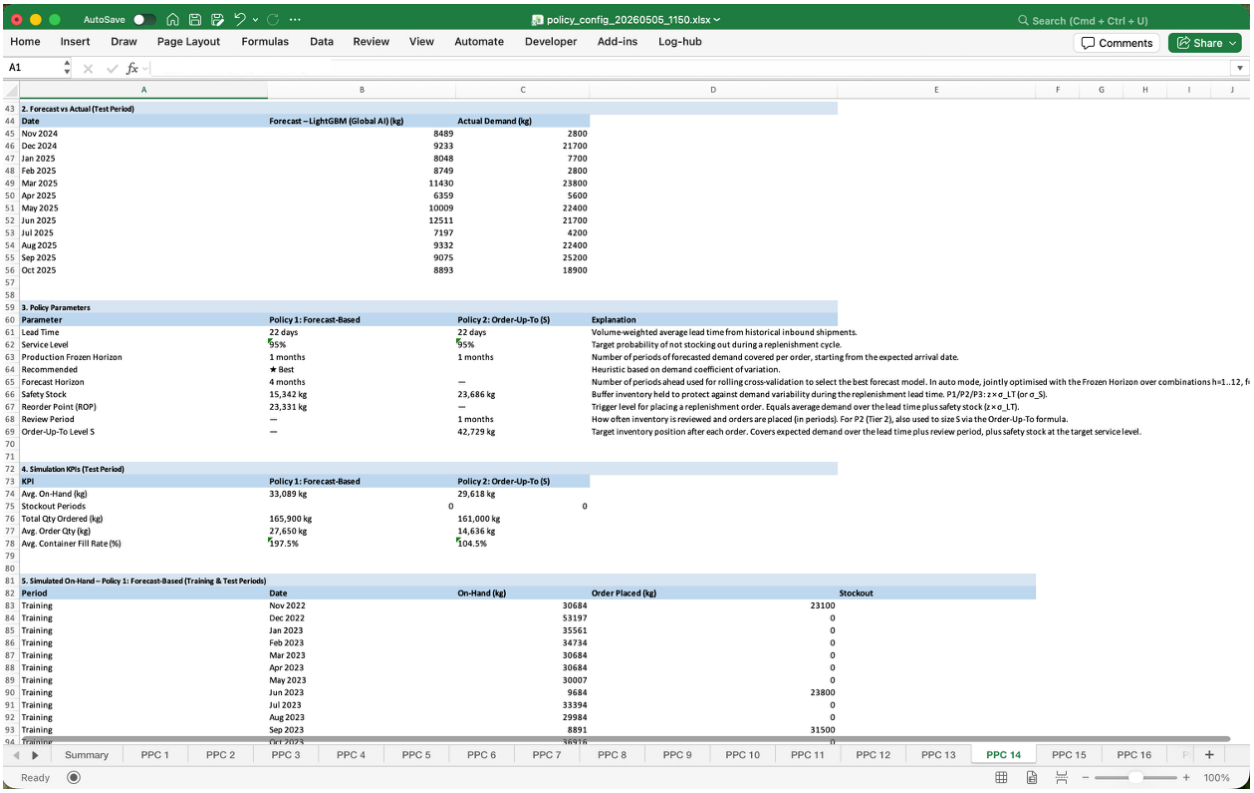


Figure 24. Inventory Policy Export: page configuration with auto/manual modes.

When exporting all materials for a specific Tier 2 plant, the page additionally offers a container fill-rate optimization for Policy 2. Materials whose individual average order already fills 85% or more of a full container retain their individually optimized review period; the remaining small-order materials are grouped under a common review period R* chosen to maximize their combined container fill rate, synchronizing ordering rhythms across materials at the same destination plant and reducing transport cost per kilogram shipped. The feature is exposed only at Tier 2 plant level, since Tier 1 replenishment is governed by production campaign economics rather than inbound container logistics.



2. Forecast vs Actual (Test Period)				
Date	Forecast – LightGBM (Global AI) (kg)	Actual Demand (kg)		
Nov 2024	8489	2800		
Dec 2024	9233	21700		
Jan 2025	8048	7700		
Feb 2025	8749	2800		
Mar 2025	11430	23800		
Apr 2025	6359	5600		
May 2025	10009	22400		
Jun 2025	12511	21700		
Jul 2025	7197	4200		
Aug 2025	9332	22400		
Sep 2025	9075	25200		
Oct 2025	8893	18900		
3. Policy Parameters				
Parameter	Policy 1: Forecast-Based	Policy 2: Order-Up-To (S)	Explanation	
Lead Time	22 days	22 days	Volume-weighted average lead time from historical inbound shipments.	
Service Level	95%	95%	Target probability of not stocking out during a replenishment cycle.	
Production Frozen Horizon	1 months	1 months	Number of periods of forecasted demand covered per order, starting from the expected arrival date.	
Recommended	★ Best		Heuristic based on demand coefficient of variation.	
Forecast Horizon	4 months	—	Number of periods ahead used for rolling cross-validation to select the best forecast model. In auto mode, jointly optimised with the Frozen Horizon over combinations h=1..12, f	
Safety Stock	15,342 kg	23,686 kg	Buffer inventory held to protect against demand variability during the replenishment lead time. P1/P2/P3: z×σ_LT (or σ_S).	
Reorder Point (ROP)	23,331 kg	—	Trigger level for placing a replenishment order. Equals average demand over the lead time plus safety stock (z×σ_LT).	
Review Period	—	1 months	How often inventory is reviewed and orders are placed (in periods). For P2 (Tier 2), also used to size S via the Order-Up-To formula.	
Order-Up-To Level S	—	42,729 kg	Target inventory position after each order. Covers expected demand over the lead time plus review period, plus safety stock at the target service level.	
4. Simulation KPIs (Test Period)				
KPI	Policy 1: Forecast-Based	Policy 2: Order-Up-To (S)		
Avg. On-Hand (kg)	33,089 kg	29,618 kg		
Stockout Periods	0	0		
Total Qty Ordered (kg)	165,900 kg	161,000 kg		
Avg. Order Qty (kg)	27,650 kg	14,636 kg		
Avg. Container Fill Rate (%)	197.5%	104.5%		
5. Simulated On-Hand – Policy 1: Forecast-Based (Training & Test Periods)				
Period	Date	On-Hand (kg)	Order Placed (kg)	Stockout
Training	Nov 2022	30684	23100	
Training	Dec 2022	53197	0	
Training	Jan 2023	35561	0	
Training	Feb 2023	34734	0	
Training	Mar 2023	30684	0	
Training	Apr 2023	30684	0	
Training	May 2023	30007	0	
Training	Jun 2023	9684	23800	
Training	Jul 2023	33394	0	
Training	Aug 2023	29984	0	
Training	Sep 2023	8891	31500	

Figure 25. Inventory Policy Export: sample workbook output.